

# Marriage Squeeze and Education Choice: Evidence from China's Imbalanced Sex Ratios

Shuning Li\*

August 8, 2026

## Abstract

China's male-biased sex ratios transformed a demographic imbalance into marriage-market competition. This paper asks whether exposure to higher male-to-female sex ratios creates incentives to acquire education before marriage. Using 2010 China census microdata for cohorts born between 1975 and 1990, the paper assigns sex ratios at birth measured at the birth-market level. Higher sex-ratio exposure is associated with later first marriage for both men and women, and with a larger delay and a wider spousal age gap for men. Education responses are positive but modest in the pooled sample and mask substantial heterogeneity: they are stronger in urban markets and in East/Northeast and Central provinces. The results remain robust after controlling for One-Child Policy intensity. Although both men and women increase education in response to higher sex ratios, they appear to do so under different incentives: men benefit more through improved matching prospects in a male-surplus market, while women benefit more through match quality and household bargaining position. A DiNardo-Fortin-Lemieux decomposition shows that education upgrading explains a larger share of the overall change in age at first marriage, while sex-ratio change contributes more for men and urban residents. The findings suggest that marriage-market pressure can create an additional incentive for schooling, but only where education is a feasible and valued margin of adjustment.

---

\*Department of Economics, University of Western Ontario. Email: [sli2744@uwo.ca](mailto:sli2744@uwo.ca). I am grateful to Audra Bowlus, Elizabeth Caucutt, Lisa Tarquinio, and Terry Sicular for their guidance, and to participants in the AMM Reading Group and at the CEA conference for helpful comments and suggestions. All errors are my own.

# 1 Introduction

China’s male-to-female sex ratio at birth rose sharply after 1980, from about 106 boys per 100 girls in 1970 to about 117 in 2010. The increase is commonly linked to the interaction of son preference, fertility restrictions under the One-Child Policy, and diffusion of sex-selective technology (Golley and Tyers, 2012; Chen and Zhang, 2019; Guilimoto, 2012). When these cohorts reached marriageable ages, the male-biased demographic imbalance became a marriage-market imbalance: men faced more competition for relatively fewer potential female spouses. A marriage squeeze is a demographic imbalance in the number of eligible men and women seeking partners.

A growing literature shows that this marriage squeeze can change pre-marital investment. In high-sex-ratio markets, families may save more, invest in housing, or accumulate other assets that improve marriage prospects (Wei and Zhang, 2011; Zhou and Wang, 2024). Education may be another margin of adjustment. It can signal ability, family background, permanent income, or social status. It can also affect match quality because marriage markets display positive assortative mating on education (Becker, 1973; Greenwood et al., 2014; Dong and Xie, 2023). If education improves marriage prospects, forward-looking individuals may respond to a tighter marriage market by investing more in schooling.

Over the same period, educational attainment increased rapidly for these cohorts. Existing work emphasizes education reforms, economic development, and quantity-quality responses to fertility policy as important drivers of rising schooling wide in China (Wu, 2010; Guo et al., 2019; Rosenzweig and Zhang, 2009; Li and Zhang, 2017). This paper studies a complementary demand-side channel. Can marriage-market pressure generated by imbalanced sex ratios create additional incentives to invest in education?

I study this question using 2010 China census microdata for cohorts born between 1975 and 1990. I assign individuals to their male-to-female sex ratio at birth (SRB) measured at the province-by-urban/rural birth-market level. This SRB measure is predetermined relative to adult marriage outcomes and education decisions. The baseline specification compares cohorts within the same birth market, controlling for birth-market fixed effects, cohort fixed effects, and individual controls.

The empirical analysis proceeds in four steps. I first document whether higher male-to-female sex ratios are reflected in marriage squeeze, when the number of marriageable men exceeds the number of marriageable women in the marriage market. A higher sex ratio is associated with later marriage for both men and women, with larger delays for men. For men, a tighter marriage market also increases the spouse age gap. These patterns are consistent with a marriage squeeze that men in more male-biased markets primarily adjust by delaying

marriage rather than remaining unmarried.

Second, I examine whether education responds to marriage-market pressure. In the full sample, the estimated education effects are positive but small. However, these average effects mask substantial heterogeneity across regions. Higher sex-ratio exposure is positively associated with the some-college-or-above margin in urban and East/Northeast markets, while the estimates are weaker or negative in rural and Western markets. This pattern suggests that the education response to marriage-market pressure depends on local market conditions, appearing where schooling is more feasible and more likely to be valued.

Third, to interpret these patterns, I study whether education has marriage-market returns through matching and bargaining margins. Education is strongly associated with spousal education, and in high-sex-ratio markets it partly offsets marriage delays for men. The bargaining evidence is more suggestive: using household headship as a proxy for intra-household status, the estimates point to stronger returns to education for women than for men. Taken together, these results suggest that education delays marriage on average, but its marriage-market return differs by gender.

Finally, I use a DiNardo-Fortin-Lemieux (DFL) decomposition to quantify the relative contribution of sex-ratio change and education upgrading to changes in age at first marriage. Education upgrading explains a larger share of the overall change in age at first marriage, while sex-ratio exposure contributes more for men than for women, and more for urban residents than for rural residents.

The paper contributes to three strands of literature. First, it contributes to work on the long-run effects of imbalanced sex ratios and marriage squeeze ([Angrist, 2002](#); [Lafortune, 2013](#); [Abramitzky et al., 2011](#); [Guilmoto, 2012](#)). The evidence directly links birth-market sex-ratio variation to marriage timing and spouse characteristics in China. Second, it contributes to work on educational expansion in China by showing that marriage-market incentives can complement education policy and fertility-policy explanations ([Rosenzweig and Zhang, 2009](#); [Wu, 2010](#); [Guo et al., 2019](#); [Li and Zhang, 2017](#)). Third, it contributes to work on pre-marital investment and marriage-market returns by separating matching and bargaining margins ([Wei and Zhang, 2011](#); [Chiappori et al., 2009](#); [Greenwood et al., 2014](#); [Iyigun and Walsh, 2007](#)).

The rest of the paper is organized as follows. Section 2 provides background on sex-ratio imbalance and education in China’s marriage market. Section 3 describes the data and construction of sex-ratio exposure. Section 4 presents the empirical strategy. Section 5 reports the main results. Section 6 studies heterogeneity. Section 7 presents robustness checks. Section 8 reports the DFL decomposition. Section 9 concludes.

## 2 Background

### 2.1 Sex-ratio imbalance and the marriage squeeze

The male-to-female sex ratio at birth is naturally close to 1.05 or 1.06 boys per girl. In China, it rose far above this biological benchmark after the introduction of the One-Child Policy (OCP).<sup>1</sup>

The One-Child Policy affected the sex ratio at birth because fertility restrictions interacted with son preference (Guilmoto, 2012; Chen and Zhang, 2019). Although enforcement allowed exemptions for some rural households, ethnic minorities, and other qualified groups, the policy substantially constrained fertility for most families. In settings where sons are preferred for patrilineal succession, old-age support, and household status, these constraints strengthened the incentive to ensure that at least one child was male. The diffusion of diagnostic ultrasound further lowered the cost of prenatal sex selection, allowing households with strong son preference to identify fetal sex and selectively abort female fetuses (Chen et al., 2013). Underreporting and delayed registration of girls may also have affected measured sex ratios, especially when households had incentives to hide daughters to preserve eligibility for another birth (Goodkind, 2011). For this reason, studies often use school-age or later-childhood cohort counts to reduce measurement error in sex ratios at birth (Wei and Zhang, 2011).

The resulting imbalance is large. Figure 1 shows the national increase in China’s sex ratio at birth relative to other countries. Figure 2 shows that the imbalance also varies across provinces and cohorts. The demographic imbalance becomes economically relevant when these cohorts enter the marriage market. A male-biased birth cohort implies that, years later, more men compete for relatively fewer women. As these cohorts age, the imbalance accumulates into a sizable surplus of men relative to women in the relevant marriage-age population. By 2020, the population of men aged 20 to 40 exceeded that of women by 17.52 million.<sup>2</sup>

---

<sup>1</sup>China began with a voluntary family-planning campaign in the early 1970s, encouraging households to limit fertility. A more restrictive One-Child Policy was piloted in 1979 and became national policy in 1980. In 1982, family planning was written into the Constitution as an obligation of Chinese citizens. Enforcement varied during the 1980s, especially in rural areas, but became more stable after 1990. The policy was implemented through fines, employment penalties, and administrative sanctions. The One-Child Policy remained in place for more than three decades before being gradually relaxed in the mid-2010s (Zhang, 2017).

<sup>2</sup>Data source from the National Bureau of Statistics

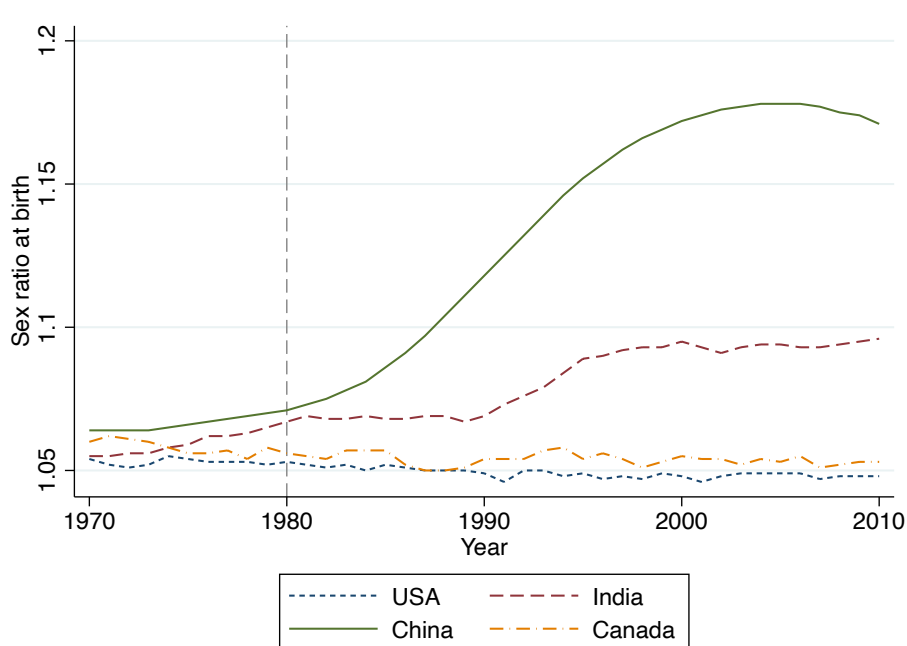


Figure 1: Sex ratio at birth across countries

*Source:* World Bank Open Data.

## 2.2 Education in the marriage market

Education is closely related to marriage-market outcomes. Figure 3 first compares never-married rates by education, gender, and age group. Among individuals aged 35–40, most marriage search has already been resolved, so remaining never married can be viewed as an equilibrium outcome of the marriage market. In this older group, men have higher never-married rates than women at each education level. The male rate is 9.1 percent among those with less than high school, 8.9 percent among high-school graduates, and 6.8 percent among those with college or above. The corresponding female rates are 2.2, 4.5, and 6.0 percent. The education gradient therefore differs by gender: lower-educated men are more likely to remain unmarried, whereas higher-educated women are more likely to remain unmarried.

The younger group shows a different pattern because many individuals aged 25–30 are still actively searching in the marriage market. For both men and women, never-married rates increase with education, consistent with the mechanical delay created by continued schooling and later labor-market entry. The level difference by gender remains large within every education category. Among individuals with less than high school, the never-married rate is 30.9 percent for men and 14.8 percent for women. Among high-school graduates, the rates are 40.3 and 25.9 percent. Among those with college or above, the rates are 54.0 and 40.3 percent. These differences are consistent with a male-biased marriage market in which

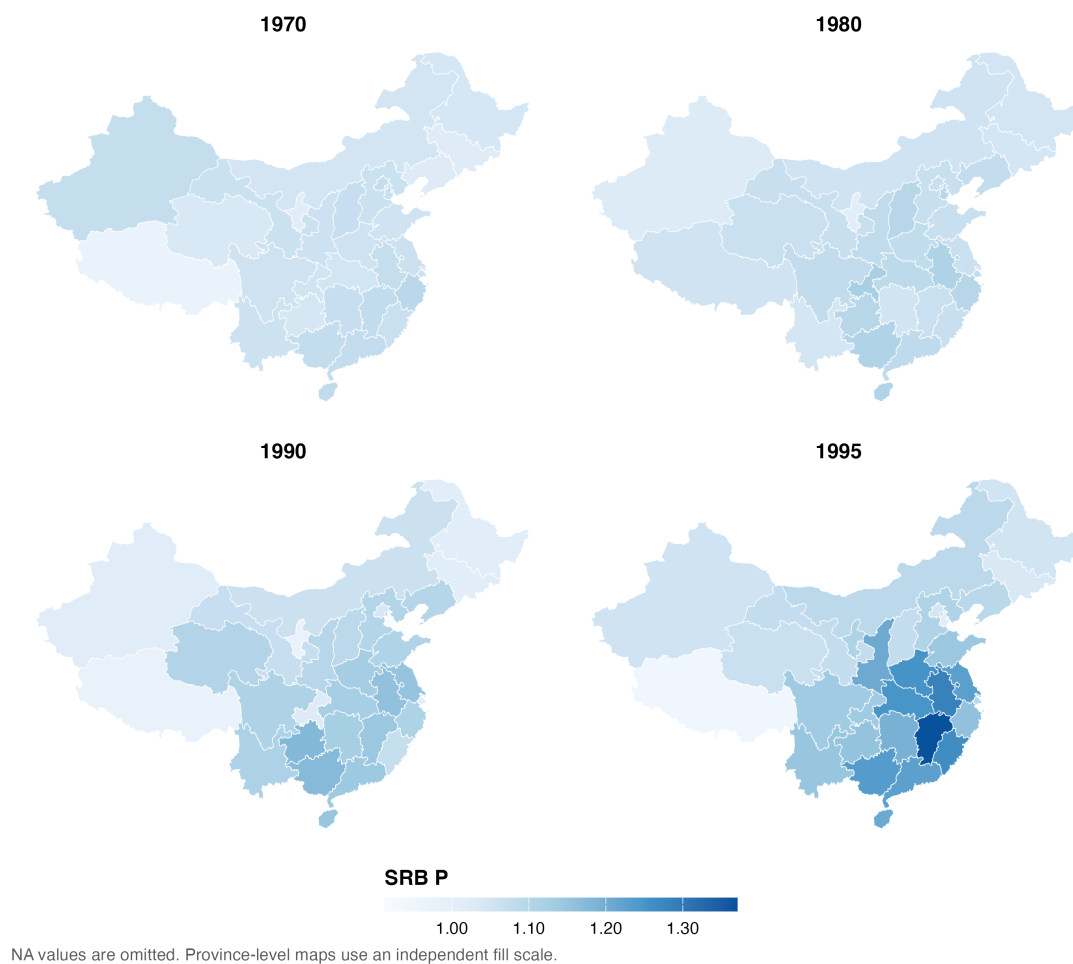


Figure 2: Province-level sex-ratio-at-birth patterns and trends

men face more difficulty marrying at a given education level, while education also delays marriage mechanically for younger cohorts.

Education also matters for whom individuals marry. Figures 4 and 5 show strong positive assortative mating on education among matched first-marriage spouses. In the 35–40 group, 87.7 percent of women with less than high school and 93.9 percent of men with less than high school marry spouses who also have less than high school. At the upper end, 80.6 percent of women with college education and 69.7 percent of men with college education marry spouses who also have college education. The pattern is at least as strong among younger cohorts: the corresponding upper-education shares rise to 83.3 percent for women and 77.8 percent for men. These descriptive patterns support the interpretation that education is a marriage-market attribute, not only a labor-market investment (Becker, 1973; Greenwood et al., 2014; Dong and Xie, 2023).

Taken together, this evidence motivates the empirical analysis below. If education affects marriage timing, the probability of remaining unmarried, and spousal education, then a tighter marriage market may change the incentive to acquire education. The key empirical question is therefore whether cohorts exposed to higher sex ratios invest more in education, and whether any education response is concentrated in markets where education has stronger marriage-market value.

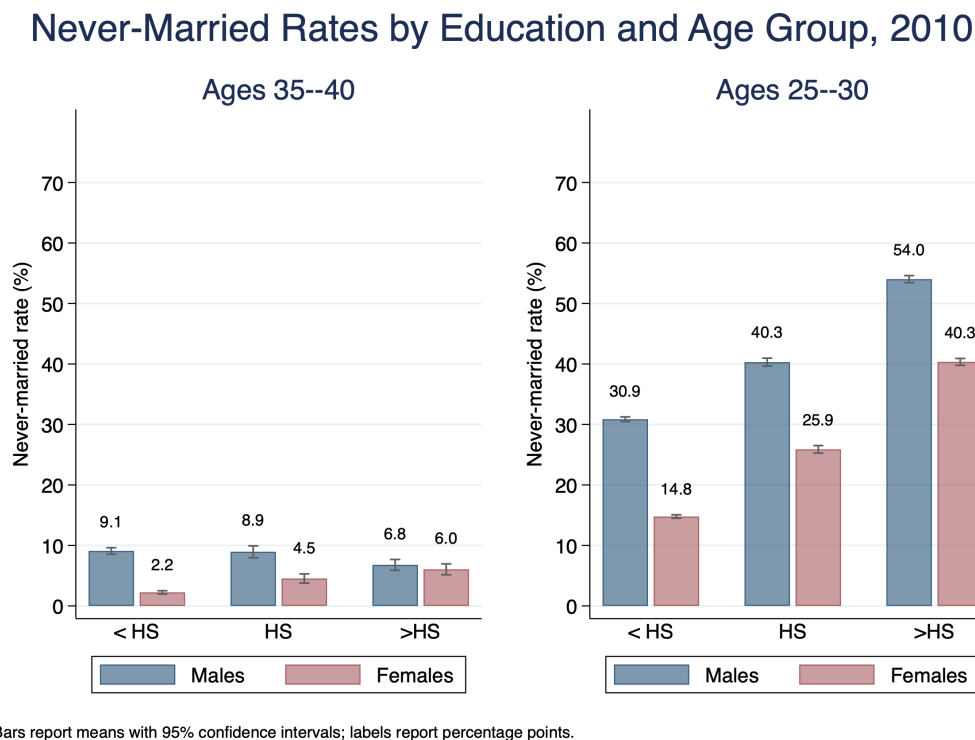


Figure 3: Never married by age and education

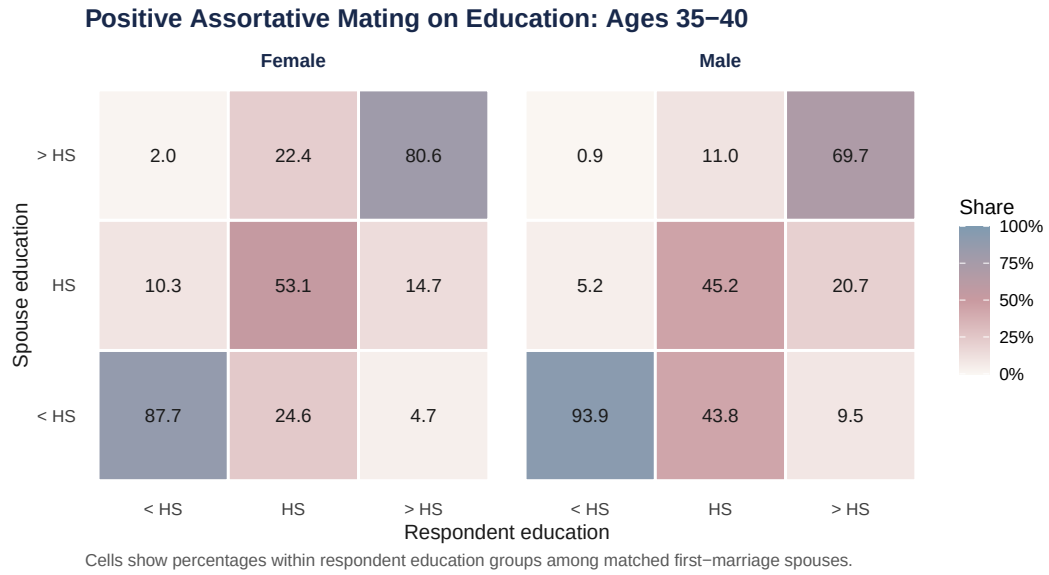


Figure 4: Assortative mating on education, older cohorts



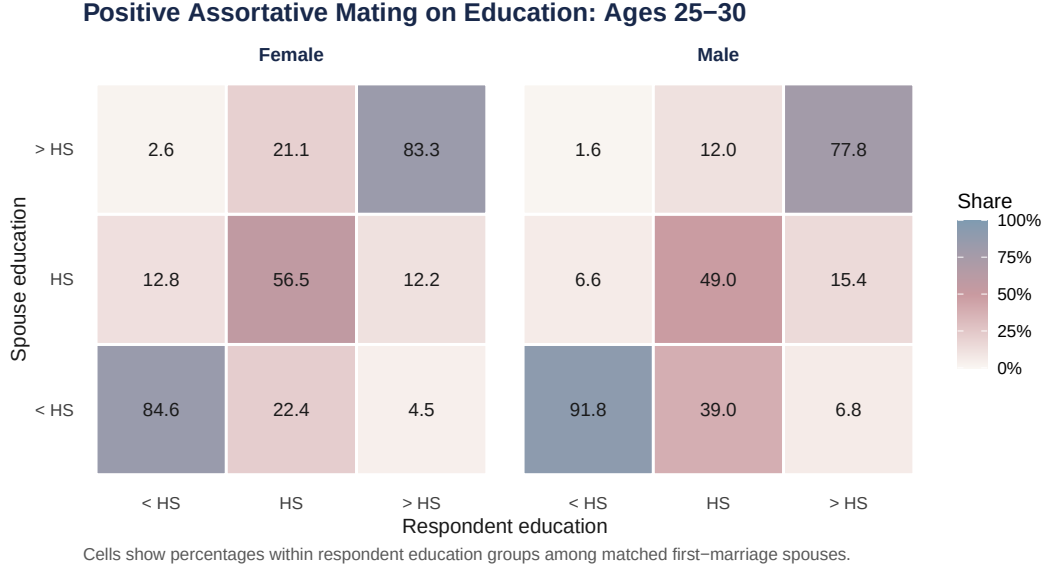


Figure 5: Assortative mating on education, younger cohorts

## 3 Data

### 3.1 Sex ratio

The key explanatory variable is the sex ratio at birth (SRB), measured as the male-to-female ratio for a birth-market cohort. I construct the SRB panel from the 1990 and 2000 one percent census microdata samples by counting males and females within each birth market and then forming a seven-year rolling cohort ratio. Appendix A describes the census sources and preprocessing procedures. I use sex ratio at birth for two reasons. First, it captures the demographic tightness that cohorts are likely to perceive as they approach the marriage market. Households do not need to know the exact local sex ratio to respond to a male surplus: information can be transmitted through social networks, marriage payments, and observed matching difficulties as cohorts grow up (Wei and Zhang, 2011). Second, birth-market sex ratios are predetermined relative to individuals' later education, marriage, and migration decisions. In contrast, contemporaneous adult sex ratios may be affected by marriage-driven or education-driven migration, making them endogenous to the outcomes of interest. Using SRB, therefore, provides a cleaner measure of exposure to demographic imbalance.

The birth market is defined at the province-by-urban/rural level. Province captures geographic variation in demographic imbalance, economic development, and marriage norms. The urban-rural dimension further separates markets that differ in schooling opportunities, income, migration, hukou status, and local matching networks. The ideal measure would

combine province of birth with urban/rural status at birth. The census reports province of birth but not urban/rural birth status, so I use current urban/rural residence to proxy for this dimension of the birth market. This choice is transparent but imperfect: rural-to-urban migration and urban reclassification can assign some rural-origin respondents to urban markets. I therefore treat urban/rural heterogeneity as an inferred-market analysis and use hukou-based and migration-related checks to assess the concern. These checks do not materially affect the results.<sup>3</sup> This definition is also consistent with observed marriage-market segmentation: among matched couples in the 2010 analysis sample, 91.8 percent share the same birth province and 93.5 percent share the same hukou status.<sup>4</sup> With 31 provinces and two urban/rural sectors, the design implies 62 birth markets.

For each cohort, I construct sex-ratio exposure using a seven-year rolling window. Individuals do not compete only with people born in the same year, but also with nearby cohorts. In the 2010 census sample, the average age gap between spouses is about two years, suggesting that the relevant marriage-market cohort pool extends beyond a single birth year. I therefore include cohorts born within three years of the respondent's birth year to capture adjacent cohorts. For example, for a person born in 1985, the sex-ratio exposure is the male-to-female ratio among individuals born between 1982 and 1988 in the same province-by-urban/rural birth market.

A limitation is that I cannot observe literal sex ratios at birth for every cohort-market cell. The ideal data would record complete birth registrations by sex, province, urban/rural origin, and cohort. The available microdata instead provide large census samples at discrete census years. I therefore approximate birth-cohort sex ratios from the 1990 and 2000 census samples, using the nearest post-birth census in which the relevant cohorts can be observed. For some cohorts the measure is based on childhood or school-age cohort counts rather than administrative birth records. This approach also helps address measurement concerns emphasized in the literature. Female births may be underreported in infancy when households face fertility restrictions, and delayed registration of girls can distort measured sex ratios at birth (Wei and Zhang, 2011). Counts observed at school age are less likely to miss children who were initially unreported, since children generally need to be registered to enter

---

<sup>3</sup>In the hukou-based assignment, respondents with non-agricultural hukou are assigned to urban birth markets and respondents with agricultural hukou are assigned to rural birth markets. This exercise assumes that hukou conversion remained difficult enough by 2010 for hukou type to preserve status at birth. The hukou-based estimates are stable relative to the current-residence-based estimates. The migration-related checks are discussed in the heterogeneity analysis.

<sup>4</sup>I report hukou-status similarity rather than urban/rural-status similarity because the matched-spouse data contain spouse hukou status but not a separate spouse-specific urban/rural residence variable. Current urban/rural residence is measured at the household level for co-resident matched couples and is therefore mechanically the same within a matched household.

school. The seven-year rolling window further smooths measurement error in small cohort-market cells. I therefore interpret the measure as early-life birth-cohort sex-ratio exposure or approximated SRB.

Table 1 relates sex-ratio exposure to One-Child Policy intensity. Cohorts born after 1980 have higher sex ratios, and higher birth fines are associated with more male-biased sex ratios. The birth-fine measure comes from [Ebenstein \(2010\)](#) and is measured in years of household income (see Appendix A.8). Market and cohort fixed effects explain about 35 percent of the variation in province-urban SRB. These patterns are consistent with the policy channel described in Section 2: fertility restrictions, when combined with son preference, increased the likelihood of male-biased birth cohorts. They also motivate the robustness checks below, which control directly for policy intensity and minority share.

Table 1: One-Child Policy and province-urban sex ratio at birth

|                          | (1)                 | (2)                 | (3)                 | (4)                 |
|--------------------------|---------------------|---------------------|---------------------|---------------------|
| Birth cohort $\geq 1980$ | 0.024***<br>(0.004) | 0.067***<br>(0.013) |                     |                     |
| Fine at birth            |                     |                     | 0.021***<br>(0.002) | 0.014***<br>(0.003) |
| Market FE                | No                  | Yes                 | No                  | Yes                 |
| Cohort FE                | No                  | Yes                 | No                  | Yes                 |
| N                        | 1,657               | 1,657               | 1,657               | 1,657               |
| R-squared                | 0.017               | 0.355               | 0.059               | 0.364               |

Notes: Dependent variable is province-urban SRB. Fine at birth is measured in years of income. Market FE are province-by-urban cells; cohort FE are birth-year fixed effects. Stars denote 10, 5, and 1 percent significance.

### 3.2 Sample selection

The individual-level analysis uses the 2010 China population census microdata. The file is a 1.58 percent sample of long-form census records. The 2010 long form was administered to 10 percent of households and contains more detailed demographic, education, household, and migration questions than the short form. Each record is equally weighted. In this paper, I use information on education, marital status, age at first marriage, migration, hukou status, household relationships, province of birth, and current residence. The analysis sample includes individuals born between 1975 and 1990, who were 20 to 35 years old in 2010. This age range focuses on cohorts that were active in the marriage market and old enough to have completed high school and, for college entrants, to have attended college.

The main marriage outcomes are never married, age at first marriage, and spouse age gap. Never married is measured from marital status in the census. Age at first marriage is observed for ever-married individuals and is therefore conditional on marriage. Spouse age gap is measured for respondents whose first-marriage spouse can be matched within the household roster.<sup>5</sup> Because the interpretation differs across outcomes, I treat never married as an extensive-margin outcome and age at first marriage as a timing outcome among those who marry.

Education is measured using a four-category schooling variable constructed from the detailed education level and the respondent’s school-completion status. The categories are less than high school, high school, some college, and BA+. Less than high school includes respondents below senior high school. High school includes senior-high graduates. Some college includes all junior-college respondents, as well as university respondents who are still enrolled, have dropped out, or otherwise have not graduated from university. BA+ includes university graduates and respondents with postgraduate education. This definition treats current postsecondary enrollment and unfinished university as part of the some-college margin. This measure is particularly important for respondents aged 20–22 in the 2010 census, many of whom are still enrolled in university. Classifying currently enrolled university students as having some college allows the analysis to retain these observations and capture the sex ratio variation among younger cohorts, while still distinguishing individuals who have entered postsecondary education from those whose highest level of schooling is high school.

Individual controls are defined from the census and external province-level data. Hukou status indicates whether the respondent holds non-agricultural hukou, and minority status indicates non-Han ethnicity. Local income is measured as log birth-market income per capita when the respondent was age 15, constructed from annual National Bureau of Statistics data. Specifically, the control is real province-by-urban/rural per-capita income in 2000 RMB, assigned by birth market and the calendar year when the respondent was age 15. It is intended to absorb local economic conditions during adolescence, when high-school continuation and postsecondary-entry decisions are likely to be made; Appendix A.9 gives the construction details.

Table 2 reports descriptive statistics by gender. The full sample contains 303,851 men and 306,057 women. Men and women are similar in age and urban residence, but differ substantially in marriage outcomes. Men have a higher never-married rate, about 0.502

---

<sup>5</sup>Matched couples are identified from household relationship codes in family households. Candidate pairs include head-spouse pairs and unambiguous child-child-in-law pairs. I retain only opposite-sex pairs with valid first-marriage information, the same reported first-marriage year, both spouses marrying at age 15 or older, and an absolute age-at-first-marriage gap no larger than 10 years. In the sample, this restriction excludes 0.83 percent of currently married respondents.

compared with 0.381 for women. Conditional on marriage, men also marry later than women: the mean age at first marriage is 24.270 for men and 22.808 for women. The education distribution also differs slightly by gender. Men have a lower share below high school and a slightly higher share with a bachelor's degree, while women have a marginally higher share in the some-college category.

The last row describes migration patterns that matter for assigning sex-ratio exposure. Birth-province migrant indicates whether the respondent's current province differs from the reported birth province. This variable does not enter the baseline exposure definition, which is based on birth province and observed 2010 urban/rural residence, but it shows that some individuals may no longer live in the market where they were exposed to early-life sex ratios. Migration therefore creates potential measurement error in treatment assignment, especially for the urban/rural component of the birth market. Ideally, I would observe childhood residence, urban/rural status at birth, complete migration histories, and the timing of migration relative to schooling. However, the 2010 census does not provide that full history, I document the issue in Appendix A.7, use migrant-type heterogeneity below, and add a direct check of whether SRB predicts cross-province migration. Excluding migrants, however, would condition on a post-exposure choice and may introduce selection bias; I therefore use migrant status mainly to interpret heterogeneity across groups.

Table 2: Descriptive statistics by gender, 2010

|                        | Males   |       | Females |       |
|------------------------|---------|-------|---------|-------|
|                        | Mean    | SD    | Mean    | SD    |
| Age                    | 26.843  | 4.725 | 26.802  | 4.689 |
| Urban                  | 0.471   | 0.499 | 0.478   | 0.500 |
| Hukou                  | 0.332   | 0.471 | 0.347   | 0.476 |
| Minority               | 0.055   | 0.228 | 0.056   | 0.230 |
| Log income             | 8.079   | 0.643 | 8.090   | 0.635 |
| Less than HS           | 0.527   | 0.499 | 0.547   | 0.498 |
| High school            | 0.192   | 0.394 | 0.171   | 0.377 |
| Some college           | 0.196   | 0.397 | 0.198   | 0.398 |
| BA+                    | 0.086   | 0.280 | 0.084   | 0.277 |
| Never married          | 0.502   | 0.500 | 0.381   | 0.486 |
| Currently married      | 0.488   | 0.500 | 0.609   | 0.488 |
| Age first marriage     | 24.270  | 3.064 | 22.808  | 2.974 |
| Birth-province migrant | 0.165   | 0.371 | 0.146   | 0.353 |
| N                      | 303,851 |       | 306,057 |       |

Notes: Sample includes individuals born between 1975 and 1990 in the 2010 population census microdata.

## 4 Estimation Strategy

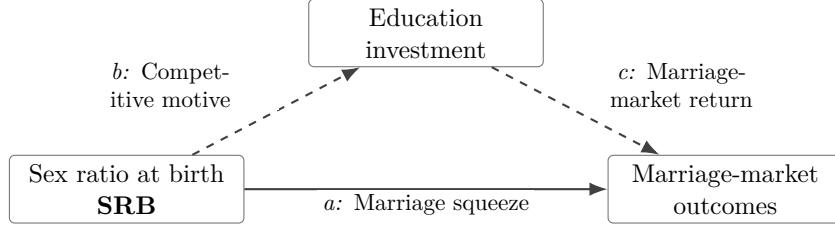


Figure 6: Conceptual framework

Figure 6 organizes three empirical paths examined in the paper. Path (a) captures the direct effect of a higher male-to-female SRB on marriage-market outcomes. A more male-biased cohort increases competition on the long side of the market and can affect marriage formation and timing through market tightness. Path (b) captures an indirect education channel. If schooling improves expected match quality, social status, or household position, individuals and families may respond to a tighter marriage market by investing more in education. Path (c) captures the marriage-market return to education itself. Education can affect marriage outcomes through matching and bargaining channels, and these education-related returns may differ in high-sex-ratio markets where the relative scarcity of men and women changes the value of marriage-market attributes.

The framework delivers three testable predictions that correspond to these paths. First, the direct marriage-squeeze path predicts that higher sex-ratio exposure should be associated with delayed or less favorable marriage-market outcomes, especially for men, who are on the long side of China’s male-biased marriage market. Second, the indirect education path predicts that higher sex-ratio exposure should be associated with greater educational attainment when schooling is a feasible and valued margin of pre-marital investment. Third, the education-return path predicts that education should be linked to marriage-market outcomes, and that the strength of this link may vary with sex-ratio exposure and gender.

The baseline specification estimates two-way fixed-effect OLS regressions separately by gender:

$$Y_{icpu} = \alpha + \beta SRB_{cpu} + \gamma X_{icpu} + \lambda_{pu} + \lambda_c + \varepsilon_{icpu}, \quad (1)$$

where  $i$  indexes individuals,  $c$  birth cohorts,  $p$  birth provinces, and  $u$  the urban/rural market.

The outcome  $Y_{icpu}$  includes both marriage outcomes and education outcomes. Marriage outcomes include an indicator for never having married, age at first marriage and spousal age gap. Education outcomes are measured using the four-category education variable. The ordered-probit specification treats the four categories as an ordered outcome. The linear probability models focus on two threshold outcomes: completing high school, and entering

postsecondary education. The first threshold captures high-school completion, while the second captures the transition into college or higher completed degrees.

The treatment variable  $SRB_{cpu}$  is standardized sex ratio at birth, so  $\beta$  is interpreted as the effect of a one-standard-deviation increase in exposure. A one-standard-deviation change is approximately 0.04 in the sex ratio, or four additional males per 100 females.

The vector  $X_{icpu}$  is predetermined individual and local controls, including hukou status, minority status, and log birth-province income per capita measured when the respondent was age 15. Hukou status captures institutional differences in schooling access, migration opportunities, and marriage markets between agricultural and non-agricultural hukou holders. Minority status accounts for differential exposure to fertility policies, as well as differences in marriage norms. Birth-province income at age 15, constructed from annual province-level data from the National Bureau of Statistics, controls for local economic conditions during the period when secondary-school decisions are made. Although market and cohort fixed effects absorb permanent market differences and national cohort shocks, this income control captures differential local economic conditions at the schooling-decision age across province-by-urban/rural markets and cohorts. When marriage outcomes are the dependent variable, I also control for the respondent's educational attainment.

All specifications include birth-market fixed effects  $\lambda_{pu}$  and birth-cohort fixed effects and  $\lambda_c$ . Birth-market fixed effects absorb persistent differences across province-by-urban/rural markets, including local marriage norms, schooling opportunities, and long-run economic conditions. Birth-cohort fixed effects absorb national trends common to a cohort, such as educational expansion and changing marriage timing. Identification therefore comes from residual changes in sex-ratio exposure across birth cohorts within the same birth market, after removing national cohort shocks and observed individual and local controls.

To study the education-return path, I estimate interaction specifications that relate marriage outcomes to education, sex-ratio exposure, and their interaction:

$$\begin{aligned} Y_{icpu} = & \alpha + \beta SRB_{cpu} + \delta_1 HS_i + \delta_2 HS_i^+ \\ & + \theta_1 (HS_i \times SRB_{cpu}) + \theta_2 (HS_i^+ \times SRB_{cpu}) + \gamma X_{icpu} \\ & + \lambda_{pu} + \lambda_c + \varepsilon_{icpu}. \end{aligned} \quad (2)$$

Here,  $HS_i$  and  $HS_i^+$  indicate the respondent's educational attainment, with less than high school as the omitted category.  $HS_i$  denotes high-school graduates, while  $HS_i^+$  includes some college and BA+. The coefficients  $\delta_1$  and  $\delta_2$  capture baseline differences in marriage outcomes by education, while  $\theta_1$  and  $\theta_2$  capture whether those education gradients vary with sex-ratio exposure. This specification links the conceptual framework to the channel analysis: the baseline SRB term corresponds to direct marriage-market pressure, the education

coefficients capture the marriage-market return to schooling, and the education-SRB interactions capture whether that return differs in more male-biased markets. Following [Lafortune \(2013\)](#), education may affect marriage-market outcomes through both matching and bargaining channels. I use age at first marriage and spouse education to capture matching-channel outcomes, and household headship as a proxy for bargaining-channel outcomes.

## 5 Results

### 5.1 Marriage squeeze

Table 3 reports the main marriage-market results. The estimates are consistent with a marriage-squeeze pattern: higher sex-ratio exposure is associated with later marriage, especially for men. A one-standard-deviation increase in SRB raises age at first marriage by 0.068 years for men and 0.029 years for women, and both estimates are statistically significant at the 1 percent level. The larger male effect is consistent with the central prediction that men, as the long side of the market, face stronger marriage-market pressure in more male-biased cohorts.

The never-married results show a different but related form of gender asymmetry. The male coefficient is close to zero and not statistically significant, while the female coefficient is negative and statistically significant. Given that the sample includes individuals ages 20 to 35, never-married status should be interpreted partly as delayed marriage rather than permanent non-marriage. The negative female coefficient therefore suggests that women exposed to higher SRB are more likely to have married by the time they are observed. By contrast, the absence of a significant male effect indicates that higher SRB does not translate into a large increase in male non-marriage in this age range.

The spouse age-gap results help explain this pattern. For men, a one-standard-deviation increase in SRB raises the spouse age gap by 0.040 years, while the corresponding estimate for women is close to zero and statistically insignificant. This suggests that men in more male-biased markets may adjust not only by marrying later, but also by matching with relatively younger wives. Such adjustment can help explain why the never-married effect for men is limited even though male marriage timing is delayed.

The education coefficients are also informative for the interpretation of marriage timing. Education is strongly associated with later marriage: relative to less than high school, HS+ raises age at first marriage by 1.751 years for men and 2.050 years for women. This pattern is important for the rest of the paper. Education delays marriage mechanically through time spent in school, but it may also matter in the marriage market through matching



and bargaining returns. The next question is therefore whether sex-ratio pressure itself is associated with educational investment.

Table 3: Marriage squeeze

|            | Never married       |                      | Age first marriage   |                      | Spouse age gap       |                      |
|------------|---------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|            | Male                | Female               | Male                 | Female               | Male                 | Female               |
| SRB        | -0.001<br>(0.001)   | -0.002**<br>(0.001)  | 0.068***<br>(0.013)  | 0.029***<br>(0.010)  | 0.040**<br>(0.016)   | -0.004<br>(0.015)    |
| Hukou      | 0.021***<br>(0.002) | -0.008***<br>(0.002) | 0.618***<br>(0.021)  | 0.539***<br>(0.019)  | -0.135***<br>(0.024) | -0.434***<br>(0.023) |
| Minority   | -0.004<br>(0.003)   | -0.017***<br>(0.003) | -0.107***<br>(0.036) | -0.201***<br>(0.030) | 0.091**<br>(0.043)   | -0.126***<br>(0.040) |
| Log income | 0.029***<br>(0.008) | 0.188***<br>(0.008)  | 0.048<br>(0.103)     | -0.147*<br>(0.089)   | -0.301***<br>(0.114) | 0.707***<br>(0.107)  |
| HS         | 0.058***<br>(0.002) | 0.121***<br>(0.002)  | 0.672***<br>(0.019)  | 0.915***<br>(0.017)  | 0.147***<br>(0.022)  | 0.209***<br>(0.022)  |
| HS+        | 0.128***<br>(0.002) | 0.244***<br>(0.002)  | 1.751***<br>(0.023)  | 2.050***<br>(0.020)  | 0.092***<br>(0.026)  | 0.595***<br>(0.026)  |
| Constant   | 0.211***<br>(0.068) | -1.228***<br>(0.063) | 23.272***<br>(0.806) | 23.303***<br>(0.702) | 3.717***<br>(0.893)  | -7.720***<br>(0.846) |
| Dep. mean  | 0.502               | 0.381                | 24.270               | 22.808               | 1.354                | -2.137               |
| N          | 303851              | 306057               | 151328               | 189521               | 105253               | 125863               |
| R-squared  | 0.464               | 0.473                | 0.288                | 0.291                | 0.061                | 0.065                |

Notes: SRB is standardized. All regressions include market and cohort fixed effects. Chow-test p-values for male-female differences are  $< 0.001$  for all outcomes.

## 5.2 Education investment

Table 4 turns to educational investment using the four-category education measure. The ordered-probit marginal effects show an upward shift in the education distribution for both men and women in response to a more male-biased SRB. A one-standard-deviation increase in SRB lowers the probability of less than high school by 0.5 percentage points for men and 0.6 percentage points for women. The probability of high school rises by 0.1 percentage points, while the probabilities of some college and BA+ each rise by about 0.2 percentage points for both genders. All four-category marginal effects are statistically significant at the 1 percent level.

Table 5 reports linear probability models for two threshold outcomes: high school or above and some college or above. The first threshold captures high-school completion, while the second captures entry into postsecondary education, including junior college, current

Table 4: Education investment, ordered-probit marginal effects

|            | Less than HS         |                      | High school          |                      | Some college         |                      | BA+                  |                      |
|------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|            | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                  |
|            | Males                | Females              | Males                | Females              | Males                | Females              | Males                | Females              |
| SRB        | -0.005***<br>(0.001) | -0.006***<br>(0.001) | 0.001***<br>(0.000)  | 0.001***<br>(0.000)  | 0.002***<br>(0.000)  | 0.002***<br>(0.000)  | 0.002***<br>(0.000)  | 0.002***<br>(0.000)  |
| Hukou      | -0.357***<br>(0.001) | -0.353***<br>(0.001) | 0.063***<br>(0.000)  | 0.058***<br>(0.000)  | 0.149***<br>(0.001)  | 0.149***<br>(0.001)  | 0.145***<br>(0.001)  | 0.146***<br>(0.001)  |
| Minority   | 0.019***<br>(0.003)  | 0.017***<br>(0.003)  | -0.003***<br>(0.001) | -0.003***<br>(0.001) | -0.008***<br>(0.001) | -0.007***<br>(0.001) | -0.008***<br>(0.001) | -0.007***<br>(0.001) |
| Log income | 0.150***<br>(0.009)  | 0.143***<br>(0.009)  | -0.026***<br>(0.002) | -0.024***<br>(0.001) | -0.063***<br>(0.004) | -0.060***<br>(0.004) | -0.061***<br>(0.004) | -0.059***<br>(0.004) |
| N          | 303,851              | 306,057              | 303,851              | 306,057              | 303,851              | 306,057              | 303,851              | 306,057              |

Notes: SRB is standardized. The dependent variable is the four-category ordered education measure. All regressions include controls as well as market and cohort fixed effects. The LR test comparing the pooled restricted ordered-probit model with gender-specific unrestricted models gives  $p < 0.001$ . The male-female difference in the SRB coefficient gives  $p = 0.340$ . Standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

university enrollment, unfinished university, and BA+. A one-standard-deviation increase in SRB raises the probability of high school or above by 0.3 percentage points for both men and women, and both estimates are statistically significant at the 1 percent level. The some-college-or-above response is close to zero: 0.1 percentage points for men and effectively zero for women. The threshold results therefore reinforce the ordered-probit evidence: higher SRB is associated with a modest upward shift in educational attainment, with the clearest average response appearing at the high-school-completion margin.

The postsecondary-entry margin combines completed schooling with ongoing enrollment and unfinished university. I therefore estimate postsecondary outcomes separately by schooling-completion status. The results suggest that in more male-biased birth markets, men are more likely to enter postsecondary education, but part of this response appears at the margin of unfinished postsecondary schooling rather than completed university degrees. The exact some-college outcome shows a similar pattern in the non-completion sample. Detailed estimates are reported in Appendix Tables 21 and 22.

Taken together, Tables 4 and 5 provide initial evidence that higher SRB is associated with a modest upward shift in educational attainment. The pooled effect is clearest at the high-school threshold, while the postsecondary margin depends on schooling status and local market conditions. These patterns motivate the next parts of the analysis: whether education has marriage-market value, whether education-related marriage outcomes vary with sex-ratio exposure, and whether the education response is concentrated in markets where schooling is a feasible and valued margin of adjustment.

Table 5: Education investment, threshold outcomes

|            | HS or above          |                      | Some college or above |                      |
|------------|----------------------|----------------------|-----------------------|----------------------|
|            | (1)<br>Males         | (2)<br>Females       | (3)<br>Males          | (4)<br>Females       |
| SRB        | 0.003***<br>(0.001)  | 0.003***<br>(0.001)  | 0.001<br>(0.001)      | -0.000<br>(0.001)    |
| Hukou      | 0.418***<br>(0.002)  | 0.434***<br>(0.002)  | 0.402***<br>(0.002)   | 0.403***<br>(0.002)  |
| Minority   | -0.026***<br>(0.003) | -0.026***<br>(0.003) | -0.001<br>(0.003)     | 0.000<br>(0.003)     |
| Log income | 0.032***<br>(0.010)  | 0.065***<br>(0.009)  | 0.149***<br>(0.009)   | 0.226***<br>(0.009)  |
| Constant   | 0.080<br>(0.078)     | -0.225***<br>(0.075) | -1.059***<br>(0.072)  | -1.683***<br>(0.071) |
| Controls   | Yes                  | Yes                  | Yes                   | Yes                  |
| Dep. mean  | 0.473                | 0.453                | 0.281                 | 0.283                |
| N          | 303,851              | 306,057              | 303,851               | 306,057              |
| R-squared  | 0.324                | 0.366                | 0.304                 | 0.331                |

Notes: SRB is standardized. Threshold outcomes are high school or above and some college or above. All regressions include controls as well as market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

### 5.3 Matching channel

The preceding results establish the first two paths in the framework: higher sex-ratio exposure is associated with marriage-market delay and, in some settings, higher educational attainment. I next examine the third path, the marriage-market return to education. Following [Lafortune \(2013\)](#), education may affect marriage-market outcomes through both matching and bargaining channels. The matching channel captures whether education helps individuals form better or faster matches in the marriage market, while the bargaining channel captures whether education is associated with stronger intra-household position after marriage. I use age at first marriage and spouse education as matching-channel outcomes, and household headship as a proxy for bargaining position. The education-SRB interactions show whether these education-related marriage-market returns differ in more male-biased markets.

Table 6 studies whether education changes the relationship between SRB and marriage outcomes. Education itself is associated with later marriage and higher never-married rates, which is expected because schooling delays labor-market entry and marriage. Relative to less than high school, HS+ raises men’s age at first marriage by 1.718 years and women’s by 2.028 years.

The interaction terms show that the marriage delay associated with higher education is smaller in high-sex-ratio markets. For age at first marriage, the high-school interaction is -0.115 for men and -0.160 for women; the HS+ interaction is -0.151 for men and -0.149 for women. These estimates suggest that education partly offsets marriage delay when sex-ratio pressure is high. Put differently, schooling delays marriage on average, but in tighter marriage markets education appears to provide a relative advantage in marriage timing.

Table 7 examines spouse education. The strongest result is positive assortative mating. High-school and HS+ education are strongly associated with having a more educated spouse. HS+ men are 63.9 percentage points more likely to have an HS+ spouse, and HS+ women are 69.5 percentage points more likely to have an HS+ spouse. The SRB main effects and SRB interactions are mostly small. This qualifies the matching-channel interpretation: sex-ratio exposure does not appear to shift spouse education much conditional on marriage, but education itself is a powerful predictor of match quality.

### 5.4 Bargaining channel

Table 8 studies household headship as a proxy for intra-household bargaining position. In the Chinese census, each household reports a household head. This designation is not a direct measure of decision-making authority, but it records a recognized position within

Table 6: Matching channel: education-SRB interactions

|                  | Never married        |                      | Age at first marriage |                      | Spouse age gap       |                      |
|------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|----------------------|
|                  | Males                | Females              | Males                 | Females              | Males                | Females              |
| SRB              | -0.012***<br>(0.001) | -0.022***<br>(0.001) | 0.130***<br>(0.016)   | 0.092***<br>(0.012)  | 0.109***<br>(0.020)  | 0.008<br>(0.017)     |
| HS               | 0.058***<br>(0.002)  | 0.119***<br>(0.002)  | 0.649***<br>(0.019)   | 0.892***<br>(0.017)  | 0.127***<br>(0.022)  | 0.204***<br>(0.023)  |
| HS+              | 0.127***<br>(0.002)  | 0.241***<br>(0.002)  | 1.718***<br>(0.023)   | 2.028***<br>(0.020)  | 0.045*<br>(0.027)    | 0.591***<br>(0.026)  |
| HS $\times$ SRB  | 0.015***<br>(0.002)  | 0.030***<br>(0.002)  | -0.115***<br>(0.024)  | -0.160***<br>(0.020) | -0.087***<br>(0.027) | -0.030<br>(0.027)    |
| HS+ $\times$ SRB | 0.016***<br>(0.001)  | 0.030***<br>(0.002)  | -0.151***<br>(0.025)  | -0.149***<br>(0.021) | -0.193***<br>(0.028) | -0.025<br>(0.027)    |
| Constant         | 0.254***<br>(0.068)  | -1.151***<br>(0.063) | 23.331***<br>(0.806)  | 23.180***<br>(0.702) | 3.817***<br>(0.893)  | -7.745***<br>(0.846) |
| Controls         | Yes                  | Yes                  | Yes                   | Yes                  | Yes                  | Yes                  |
| Dep. mean        | 0.502                | 0.381                | 24.270                | 22.808               | 1.354                | -2.137               |
| N                | 303851               | 306057               | 151328                | 189521               | 105253               | 125863               |
| R-squared        | 0.464                | 0.474                | 0.288                 | 0.291                | 0.061                | 0.065                |

Notes: All regressions include market and cohort fixed effects. SSR-based Chow p-values comparing male and female full interaction specifications are  $< 0.001$  for never married, age at first marriage, and spouse age gap.

Table 7: Matching channel: spouse education

|                  | Spouse HS+          |                     | Spouse HS+          |                     |
|------------------|---------------------|---------------------|---------------------|---------------------|
|                  | Males               | Females             | Males               | Females             |
| SRB              | -0.002<br>(0.003)   | 0.002<br>(0.002)    | 0.003*<br>(0.002)   | -0.000<br>(0.001)   |
| HS               | 0.420***<br>(0.004) | 0.501***<br>(0.004) | 0.051***<br>(0.002) | 0.127***<br>(0.003) |
| HS+              | 0.659***<br>(0.004) | 0.615***<br>(0.004) | 0.639***<br>(0.004) | 0.695***<br>(0.004) |
| HS $\times$ SRB  | 0.001<br>(0.005)    | 0.002<br>(0.004)    | 0.004<br>(0.003)    | 0.001<br>(0.004)    |
| HS+ $\times$ SRB | 0.001<br>(0.003)    | -0.003<br>(0.003)   | 0.002<br>(0.004)    | 0.003<br>(0.003)    |
| Constant         | 0.008<br>(0.104)    | 0.128<br>(0.103)    | -0.117<br>(0.082)   | -0.123<br>(0.078)   |
| Controls         | Yes                 | Yes                 | Yes                 | Yes                 |
| Dep. mean        | 0.351               | 0.396               | 0.183               | 0.202               |
| N                | 105253              | 125863              | 105253              | 125863              |
| R-squared        | 0.572               | 0.539               | 0.585               | 0.585               |

Notes: SRB is standardized. All regressions include market and cohort fixed effects. SSR-based Chow p-values comparing male and female full interaction specifications are  $< 0.001$  for spouse HS+ and spouse HS+.

the household and is one of the few census measures that can speak to intra-household status. The analysis is restricted to family households, excluding collective households such as military units, university dormitories, and other institutional living arrangements. This restriction keeps the outcome closer to family organization rather than institutional residence.

The bargaining interpretation is strongest for currently married couples. Within marriage, being recorded as the household head can reflect relative position within the couple, so the currently married sample provides the main test for this channel. For never-married respondents, headship has a more mixed interpretation because it can reflect both independent household formation and status within a multi-person family household.

Table 8: Bargaining channel by marital status: household headship

|                  | Currently married    |                      | Divorced/widowed  |                     | Never married        |                      |
|------------------|----------------------|----------------------|-------------------|---------------------|----------------------|----------------------|
|                  | (1)                  | (2)                  | (3)               | (4)                 | (5)                  | (6)                  |
|                  | Males                | Females              | Males             | Females             | Males                | Females              |
| SRB              | -0.008**<br>(0.003)  | 0.008***<br>(0.001)  | 0.014<br>(0.022)  | 0.002<br>(0.019)    | 0.008***<br>(0.002)  | 0.011***<br>(0.002)  |
| HS               | -0.026***<br>(0.004) | 0.020***<br>(0.002)  | -0.001<br>(0.028) | 0.012<br>(0.026)    | 0.009***<br>(0.003)  | 0.030***<br>(0.003)  |
| HS+              | 0.026***<br>(0.005)  | 0.064***<br>(0.003)  | 0.078*<br>(0.041) | 0.138***<br>(0.030) | 0.028***<br>(0.003)  | 0.020***<br>(0.003)  |
| HS $\times$ SRB  | 0.013***<br>(0.005)  | -0.012***<br>(0.003) | 0.026<br>(0.035)  | -0.019<br>(0.026)   | -0.008***<br>(0.003) | -0.008**<br>(0.003)  |
| HS+ $\times$ SRB | 0.015***<br>(0.005)  | -0.016***<br>(0.003) | -0.002<br>(0.045) | 0.037<br>(0.029)    | -0.020***<br>(0.002) | -0.017***<br>(0.003) |
| Constant         | -0.616***<br>(0.143) | 0.233***<br>(0.072)  | -0.400<br>(0.947) | 0.848<br>(0.911)    | 1.624***<br>(0.154)  | 2.010***<br>(0.170)  |
| Controls         | Yes                  | Yes                  | Yes               | Yes                 | Yes                  | Yes                  |
| Dep. mean        | 0.500                | 0.088                | 0.379             | 0.372               | 0.150                | 0.109                |
| N                | 141,411              | 182,283              | 2,969             | 3,084               | 112,100              | 83,977               |
| R-squared        | 0.092                | 0.062                | 0.099             | 0.176               | 0.117                | 0.120                |

Notes: SRB is standardized. All regressions include market and cohort fixed effects. Controls are non-agricultural hukou, minority status, and log birth-province income per capita at age 15. Chow-test p-values for male-female differences are  $< 0.001$  for the currently-married and never-married samples. Standard errors are in parentheses.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Columns 1 and 2 report the currently married sample. The level difference in headship is large: 50.0 percent of currently married men are recorded as household heads, compared with only 8.8 percent of currently married women. Against this low female baseline, the SRB estimate is economically meaningful. For currently married women in the omitted less-than-high-school group, a one-standard-deviation increase in SRB is associated with a 0.8

percentage point increase in the probability of being recorded as household head, which is about 9 percent of the female mean. The corresponding estimate for currently married men is negative, at -0.8 percentage points. This male-female difference is statistically significant at conventional levels. I interpret this pattern as suggestive evidence that high-SRB markets shift relative household position within marriage: married men are less likely to be recorded as household heads, while married women are more likely to be recorded as household heads.

The divorced/widowed sample in columns 3 and 4 provides less informative evidence for this channel. The sample is much smaller than the currently married and never-married samples, and the SRB coefficients are not statistically significant for either gender.

The never-married sample provides a useful contrast with the currently married sample. The main difference is the sign reversal for men: the SRB coefficient is positive for never-married men but negative for currently married men. For women, the coefficient remains positive in the currently married sample. This sign flip is central to the bargaining-channel interpretation. For never-married men, headship can partly reflect living arrangements, including independent household formation. Within marriage, by contrast, headship is more directly related to relative position inside the couple. The estimates therefore support a marital-status-specific interpretation: in high-SRB markets, married women appear to hold a stronger relative household position, while the headship response among never-married individuals also reflects household formation and composition.<sup>6</sup>

Education is also related to headship, especially for currently married women. Relative to less than high school, high school is associated with a 2.0 percentage point higher probability that a currently married woman is recorded as household head, and HS+ is associated with a 6.4 percentage point higher probability. For currently married men, the high-school coefficient is negative, while the HS+ coefficient is positive but smaller than the corresponding female coefficient. These gradients are consistent with education being more strongly associated with women's household status than men's. The education-SRB interactions, however, do not indicate that this bargaining return is amplified in high-SRB markets: they are negative for currently married women and positive for currently married men. I therefore interpret the table as evidence that high sex ratios are associated with women's relative household position within marriage, rather than as evidence that high sex ratios strengthen

---

<sup>6</sup>Among never-married individuals in family households, about 9.6 percent live in single-person family households and are mechanically recorded as household heads. Under the same specification, the single-person-household outcome is positively associated with SRB for both less-than-high-school men and women, with a larger coefficient for men than for women. The remaining 90.4 percent live in multi-person family households. Within this group, the headship estimates point to a conditional household-position margin: the SRB coefficient is small and statistically insignificant for men, and positive and statistically significant for women. Thus, for never-married individuals, SRB appears to affect both household composition and headship conditional on household composition.



the bargaining return to education.

## 6 Heterogeneous Effects

The modest full-sample education effects may hide stronger responses in some markets and weaker responses in others. Birth markets differ substantially in economic development, schooling opportunities, migration patterns, and marriage-market institutions, so the same increase in sex-ratio pressure need not translate into the same educational response everywhere. I therefore examine heterogeneity along four dimensions: region, urban/rural status, migrant type, and birth-province population size. These dimensions capture differences in local development, market segmentation, exposure to the birth-market sex ratio, and the scale of the marriage market. They also organize the analysis around conditions under which education is more likely to be a feasible investment and a valued marriage-market attribute.

To keep the heterogeneity tables focused, I report only the coefficient on SRB. All specifications include the same controls and fixed effects as the baseline model. Full estimates with controls are reported in Appendix Tables 23–26.

### 6.1 Birth-province region

The regional split follows the National Bureau of Statistics’ economic-region classification, which groups provinces into eastern, central, western, and northeastern China. I combine the eastern and northeastern regions in the regression because both regions had high urbanization and high population-adjusted educational attainment in 2010, even though the Northeast differs from the eastern coast in industrial history and income performance.<sup>7</sup> This split frames the heterogeneity analysis as a comparison across schooling access, urban opportunity sets, and household constraints rather than as a purely geographic exercise.

Table 9 splits the sample by birth-province region. The regional results show that the modest full-sample education effects mask substantial heterogeneity across local markets. In East/Northeast provinces, higher SRB is positively associated with both education thresholds for men and women. A one-standard-deviation increase in SRB raises the probability of high school or above by 0.5 percentage points for men and 0.7 percentage points for women, and raises the probability of some college or above by 0.6 percentage points for men and 0.7 percentage points for women. In Central provinces, the high-school margin is positive but

---

<sup>7</sup>Appendix A.2 reports the province classification, background statistics, and interpretation of the East/Northeast grouping. The grouping should be read as a high-urbanization and high-schooling-infrastructure comparison group, not as a claim that the eastern coast and the old industrial Northeast are identical.

small and not statistically significant, while the some-college-or-above margin remains positive and statistically significant for both men and women. In Western provinces, by contrast, the estimates are negative for both education margins and both genders, with especially large negative effects at the some-college-or-above margin.

The regional split is motivated by differences in schooling access, local returns to education, and household constraints. The positive estimates in East/Northeast and Central provinces are consistent with an education response concentrated in markets where schooling can serve as a feasible signal of income, status, or match quality. In Western provinces, the negative estimates indicate that this schooling margin is less visible; tighter marriage-market conditions may be reflected in other forms of investment, earlier labor-market entry, or constraints that limit additional schooling.

Table 9: Heterogeneity by birth-province region: education outcomes

|                                       | East/Northeast |          | Central  |         | West      |           |
|---------------------------------------|----------------|----------|----------|---------|-----------|-----------|
|                                       | (1)            | (2)      | (3)      | (4)     | (5)       | (6)       |
|                                       | Males          | Females  | Males    | Females | Males     | Females   |
| <i>Panel A: High school or above</i>  |                |          |          |         |           |           |
| SRB                                   | 0.005***       | 0.007*** | 0.002    | 0.003   | -0.002    | -0.005*** |
|                                       | (0.001)        | (0.001)  | (0.003)  | (0.003) | (0.002)   | (0.002)   |
| Dep. mean                             | 0.547          | 0.524    | 0.438    | 0.407   | 0.404     | 0.400     |
| N                                     | 123355         | 126417   | 96839    | 95392   | 83657     | 84248     |
| <i>Panel B: Some college or above</i> |                |          |          |         |           |           |
| SRB                                   | 0.006***       | 0.007*** | 0.007*** | 0.006** | -0.009*** | -0.014*** |
|                                       | (0.002)        | (0.001)  | (0.002)  | (0.002) | (0.002)   | (0.002)   |
| Dep. mean                             | 0.339          | 0.342    | 0.242    | 0.234   | 0.242     | 0.249     |
| N                                     | 123355         | 126417   | 96839    | 95392   | 83657     | 84248     |
| Controls                              | Yes            | Yes      | Yes      | Yes     | Yes       | Yes       |

Notes: SRB is standardized. Panel A reports high school or above; Panel B reports some college or above. All regressions control for hukou status, minority status, log birth-province income per capita at age 15, and market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Columns split by birth-province region and gender. East/Northeast combines Northeast with East.

## 6.2 Urban and rural markets

Table 10 shows a similar contrast between urban and rural markets. In urban birth markets, higher SRB is positively associated with both educational margins for men and women. The coefficients are similar across genders: about 0.5–0.6 percentage points for the high-

school margin and 0.5 percentage points for the some-college-or-above margin. In rural birth markets, the pattern reverses. The high-school estimates are -0.4 percentage points for men and -0.7 percentage points for women, while the some-college-or-above estimates are -0.9 and -1.1 percentage points. The urban-rural split is informative because these markets differ in school availability, hukou-related access, expected returns to education, and the forms of investment used in marriage competition. Because urban/rural status at birth is not observed, these estimates should be read as inferred birth-market estimates that use current urban/rural residence as the proxy; rural-to-urban migration is likely to attenuate or blur this contrast. The positive urban estimates align with a setting in which education is both accessible and visible in the marriage market. The negative rural estimates place the response outside these schooling margins, a pattern consistent with stronger resource constraints or adjustment through assets, migration, brideprice, or other family investments.

Table 10: Heterogeneity by urban status: education outcomes

|                                       | Urban               |                     | Rural                |                      |
|---------------------------------------|---------------------|---------------------|----------------------|----------------------|
|                                       | (1)<br>Males        | (2)<br>Females      | (3)<br>Males         | (4)<br>Females       |
| <i>Panel A: High school or above</i>  |                     |                     |                      |                      |
| SRB                                   | 0.005***<br>(0.001) | 0.006***<br>(0.001) | -0.004<br>(0.002)    | -0.007***<br>(0.002) |
| Dep. mean                             | 0.696               | 0.687               | 0.275                | 0.239                |
| N                                     | 143135              | 146413              | 160716               | 159644               |
| <i>Panel B: Some college or above</i> |                     |                     |                      |                      |
| SRB                                   | 0.005***<br>(0.001) | 0.005***<br>(0.001) | -0.009***<br>(0.002) | -0.011***<br>(0.002) |
| Dep. mean                             | 0.459               | 0.464               | 0.123                | 0.116                |
| N                                     | 143135              | 146413              | 160716               | 159644               |
| Controls                              | Yes                 | Yes                 | Yes                  | Yes                  |

Notes: SRB is standardized. Panel A reports high school or above; Panel B reports some college or above. All regressions control for hukou status, minority status, log birth-province income per capita at age 15, and market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Columns split by urban status and gender.

### 6.3 Migrant type

Table 11 examines heterogeneity by migrant type. I classify individuals using birth province, current province, and hukou registration location. Type 1 are birth-province stayers whose

current residence remains in the birth province and who have not left their hukou registration place. Type 2 are temporary migrants within the birth province, defined as individuals who have left their hukou registration place for more than six months but still live in their birth province. Type 3 are temporary cross-province migrants, who have left their hukou registration place for more than six months and currently live outside their birth province. Type 4 are permanent cross-province migrants, whose current province differs from their birth province and whose hukou registration has also moved.

This classification is useful because the treatment variable is assigned based on the birth market. For individuals who remain within the birth province, birth-market SRB is more likely to represent the relevant exposure. For cross-province migrants, however, the assigned treatment may be measured with error. Their later outcomes may reflect exposure to the destination province, but the census does not reveal when migration occurred. As a result, it is difficult to assign a more accurate sex-ratio treatment for these individuals.

At the same time, migration is itself a choice and may respond to education, labor-market opportunities, or marriage-market conditions. I use the migrant-type results to describe how the education response varies with likely exposure to the birth market and with mobility status. The purpose is to show whether the education response is stronger among groups for whom birth-market SRB is more likely to represent the relevant exposure. A direct migration check does not support the concern that higher-SRB birth markets systematically induce large cross-province out-migration.<sup>8</sup>

Consistent with this interpretation, the education response is most visible among individuals who remain within the birth province, especially Types 1 and 2. For stayers, higher SRB is positively associated with both education thresholds for men and with both thresholds for women. Among within-province temporary migrants, the positive response remains visible, especially for male some-college-or-above attainment and female high-school attainment. By contrast, the estimates for cross-province migrants are weaker. For Type 3 migrants, the coefficients are close to zero or negative, and for Type 4 migrants the estimates are generally not statistically significant except for HS men.

The gender pattern also echoes the full-sample results. Men’s response appears more clearly at the postsecondary-entry margin in some subsamples, while women’s response is often at the high-school margin. This suggests that the education response to sex-ratio pressure may operate through different schooling margins for men and women. Overall, the migrant-type results support the use of birth-market exposure for the main analysis, while

---

<sup>8</sup>I regress a cross-province migration indicator on standardized SRB using the same birth-market and cohort fixed effects, separately by gender. In the national sample, the SRB coefficient is small and statistically insignificant for both men and women. By birth region, only East/Northeast men show a small positive association, about 0.24 percentage points per standard deviation of SRB.

also showing that the estimated education response is strongest for individuals who remain in their birth market.

Table 11: Heterogeneity by migrant type: education outcomes

|                                       | Type 1              |                     | Type 2             |                   | Type 3            |                    | Type 4             |                  |
|---------------------------------------|---------------------|---------------------|--------------------|-------------------|-------------------|--------------------|--------------------|------------------|
|                                       | (1)                 | (2)                 | (3)                | (4)               | (5)               | (6)                | (7)                | (8)              |
|                                       | Males               | Females             | Males              | Females           | Males             | Females            | Males              | Females          |
| <i>Panel A: High school or above</i>  |                     |                     |                    |                   |                   |                    |                    |                  |
| SRB                                   | 0.006***<br>(0.002) | 0.006***<br>(0.002) | 0.002<br>(0.002)   | 0.003*<br>(0.002) | 0.001<br>(0.003)  | -0.002<br>(0.003)  | 0.009**<br>(0.004) | 0.008<br>(0.005) |
| Dep. mean                             | 0.413               | 0.401               | 0.546              | 0.512             | 0.465             | 0.439              | 0.622              | 0.568            |
| N                                     | 148451              | 144550              | 105415             | 116855            | 38911             | 34064              | 11072              | 10588            |
| <i>Panel B: Some college or above</i> |                     |                     |                    |                   |                   |                    |                    |                  |
| SRB                                   | 0.003**<br>(0.002)  | 0.002<br>(0.002)    | 0.004**<br>(0.002) | 0.001<br>(0.002)  | -0.003<br>(0.003) | -0.005*<br>(0.003) | -0.001<br>(0.004)  | 0.001<br>(0.005) |
| Dep. mean                             | 0.229               | 0.241               | 0.348              | 0.333             | 0.254             | 0.249              | 0.447              | 0.407            |
| N                                     | 148451              | 144550              | 105415             | 116855            | 38911             | 34064              | 11072              | 10588            |
| Controls                              | Yes                 | Yes                 | Yes                | Yes               | Yes               | Yes                | Yes                | Yes              |

Notes: SRB is standardized. Panel A reports high school or above; Panel B reports some college or above. All regressions control for hukou status, minority status, log birth-province income per capita at age 15, and market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Columns split by migrant type and gender. Type 1: birth-province stayers; Type 2: within-province temporary migration; Type 3: cross-province temporary migration; Type 4: permanent cross-province migration.

## 6.4 Birth-province population size

Table 12 examines whether the education response differs by birth-province population size. The results suggest that the relevant education margin varies with market size. In smaller birth provinces, higher SRB is positively associated with the high-school margin: a one-standard-deviation increase in SRB raises the probability of high school or above by 0.7 percentage points for men and 0.9 percentage points for women. The some-college-or-above margin, however, is close to zero and statistically insignificant. In larger birth provinces, the pattern shifts to the postsecondary-entry margin. Higher SRB raises the probability of some college or above by 0.7 percentage points for men and 0.8 percentage points for women, but is negatively associated with the high-school-or-above threshold.

The opposite signs on the high-school-or-above and some-college-or-above coefficients in larger provinces imply a decline in the intermediate high-school-only category. The large-province result points to a shift in the relevant schooling margin: higher SRB is associated with greater postsecondary entry, while the high-school-only category becomes smaller.

Table 12: Heterogeneity by birth-province population size: education outcomes

|                                       | Large population    |                     | Small population    |                     |
|---------------------------------------|---------------------|---------------------|---------------------|---------------------|
|                                       | (1)                 | (2)                 | (3)                 | (4)                 |
|                                       | Males               | Females             | Males               | Females             |
| <i>Panel A: High school or above</i>  |                     |                     |                     |                     |
| SRB                                   | -0.001<br>(0.002)   | -0.002<br>(0.002)   | 0.007***<br>(0.001) | 0.009***<br>(0.001) |
| Dep. mean                             | 0.475               | 0.445               | 0.471               | 0.463               |
| N                                     | 167087              | 167269              | 136764              | 138788              |
| <i>Panel B: Some college or above</i> |                     |                     |                     |                     |
| SRB                                   | 0.007***<br>(0.002) | 0.008***<br>(0.002) | 0.002<br>(0.002)    | -0.001<br>(0.001)   |
| Dep. mean                             | 0.267               | 0.262               | 0.298               | 0.307               |
| N                                     | 167087              | 167269              | 136764              | 138788              |
| Controls                              | Yes                 | Yes                 | Yes                 | Yes                 |

Notes: SRB is standardized. Panel A reports high school or above; Panel B reports some college or above. All regressions control for hukou status, minority status, log birth-province income per capita at age 15, and market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Columns split by birth-province population size and gender. Large/small split at median birth-province population.

## 7 Robustness

The prior analysis interprets sex-ratio exposure as a source of marriage-market pressure. A remaining concern arises because variation in SRB likely reflects variation in the intensity of the One-Child Policy, which could directly affect education investments through its effect on family size and resource allocation. For example, if lower fertility increases parental investment per child, the estimated SRB coefficients may partly reflect a quantity-quality channel rather than a marriage-market channel (Rosenzweig and Zhang, 2009; Li and Zhang, 2017). To assess this concern, I add two measures of fertility-policy intensity.

The first measure is the One-Child Policy birth fine from Ebenstein (2010), measured as the monetary penalty for an unauthorized birth in years of local income. This variable captures cross-province and over-time variation in the financial cost of violating fertility restrictions. The second measure is the province-cohort minority share, which I construct from the census microdata. Ethnic minorities generally faced less restrictive fertility rules, so a higher minority share proxies for weaker exposure to strict fertility limits within a province-cohort cell.

Table 13 reports the robustness check for age at first marriage. Tables 14 and 15 report birth-region robustness checks for education outcomes. The full-sample education estimates are stable as well, but I keep the regional robustness tables in the main text because the substantive education result is the regional heterogeneity rather than the small pooled average.<sup>9</sup> These controls are not intended to fully separate all channels through which the One-Child Policy affected households. Instead, they test whether the SRB estimates are absorbed once direct measures of policy intensity are included.

The age-at-first-marriage results are very stable. The male SRB coefficient is 0.068 in the baseline specification, 0.067 after controlling for birth fines, and 0.067 after controlling for province-cohort minority share. The female coefficient remains between 0.029 and 0.030 across specifications. This stability suggests that the marriage-timing effect of SRB is not absorbed by observable measures of fertility-policy intensity.

The policy controls are themselves informative. Higher birth fines are associated with later marriage for men and earlier marriage for women: the fine coefficient is 0.096 for men and -0.063 for women, both statistically significant at the 1 percent level. This gender-asymmetric pattern is consistent with the idea that stricter fertility policy may have affected marriage timing through channels beyond the sex-ratio measure, including family-size constraints. Province-cohort minority share is negatively associated with male age at first mar-

---

<sup>9</sup>In the full-sample robustness tables, the high-school-or-above SRB coefficient remains about 0.1–0.2 percentage points across baseline, fine-control, and minority-share specifications, while the some-college-or-above coefficient remains close to zero.

riage and not significantly related to female age at first marriage. Since minority populations generally faced less restrictive fertility rules and may also differ in marriage norms, this coefficient should be interpreted as capturing a broader set of province-cohort differences rather than a pure policy effect. Importantly, including either control leaves the SRB coefficients almost unchanged.

Table 13: Robustness checks: age at first marriage

|                         | Baseline             |                      | Fine                 |                      | Minority province    |                      |
|-------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                         | (1)<br>Male          | (2)<br>Female        | (3)<br>Male          | (4)<br>Female        | (5)<br>Male          | (6)<br>Female        |
| SRB                     | 0.068***<br>(0.013)  | 0.029***<br>(0.010)  | 0.067***<br>(0.013)  | 0.030***<br>(0.010)  | 0.067***<br>(0.013)  | 0.029***<br>(0.010)  |
| Hukou                   | 0.618***<br>(0.021)  | 0.539***<br>(0.019)  | 0.618***<br>(0.021)  | 0.539***<br>(0.019)  | 0.618***<br>(0.021)  | 0.539***<br>(0.019)  |
| Minority                | -0.107***<br>(0.036) | -0.201***<br>(0.030) | -0.107***<br>(0.036) | -0.201***<br>(0.030) | -0.104***<br>(0.036) | -0.202***<br>(0.030) |
| Log income              | 0.048<br>(0.103)     | -0.147*<br>(0.089)   | 0.061<br>(0.103)     | -0.155*<br>(0.089)   | 0.044<br>(0.103)     | -0.147*<br>(0.089)   |
| HS                      | 0.672***<br>(0.019)  | 0.915***<br>(0.017)  | 0.673***<br>(0.019)  | 0.914***<br>(0.017)  | 0.672***<br>(0.019)  | 0.915***<br>(0.017)  |
| HS+                     | 1.751***<br>(0.023)  | 2.050***<br>(0.020)  | 1.751***<br>(0.023)  | 2.049***<br>(0.020)  | 1.750***<br>(0.023)  | 2.050***<br>(0.020)  |
| Fine                    |                      |                      | 0.096***<br>(0.032)  | -0.063***<br>(0.023) |                      |                      |
| Minority share province |                      |                      |                      |                      | -0.293*<br>(0.160)   | 0.032<br>(0.134)     |
| Constant                | 23.272***<br>(0.806) | 23.303***<br>(0.702) | 23.105***<br>(0.807) | 23.410***<br>(0.702) | 23.316***<br>(0.806) | 23.299***<br>(0.702) |
| Dep. mean               | 24.270               | 22.808               | 24.270               | 22.808               | 24.270               | 22.808               |
| N                       | 151328               | 189521               | 151328               | 189521               | 151328               | 189521               |
| R-squared               | 0.288                | 0.291                | 0.288                | 0.291                | 0.288                | 0.291                |

Notes: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . All regressions include market and cohort fixed effects. Fine is the OCP birth fine in years of income. Minority share province is measured at the province-cohort level.

Tables 14 and 15 repeat the regional heterogeneity analysis after adding the OCP birth fine and province-cohort minority share. Each table first reports the baseline regional estimates and then reports estimates that add both policy controls for the same threshold outcome.

The main regional pattern is largely unchanged. For the high-school margin, the SRB coefficients remain positive in East/Northeast provinces, at 0.5 percentage points for both



men and women after adding policy controls, small and statistically insignificant in Central provinces, and negative in Western provinces. For the some-college-or-above margin, the positive coefficients in East/Northeast and Central provinces remain stable, at 0.6–0.7 percentage points, while the Western coefficients remain negative at -0.9 percentage points for men and -1.4 percentage points for women. These results show that the regional heterogeneity in education responses is not reduced by direct controls for OCP intensity.

The policy controls are also related to education outcomes. Higher birth fines, which indicate stricter OCP enforcement, are positively associated with some female education margins in East/Northeast China, while minority share often enters negatively. This pattern is consistent with a quantity-quality channel in which stronger fertility restrictions increase parental investment per child, although these controls may also capture differences in local norms, geography, and development. Importantly, adding these controls leaves the SRB coefficients close to the baseline regional estimates.

Table 14: Robustness checks for education outcome: high school or above, by birth region

|                                                                        | East/Northeast       |                      | Central              |                      | West                 |                      |
|------------------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                                                        | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|                                                                        | Males                | Females              | Males                | Females              | Males                | Females              |
| <i>Panel A: Baseline: High school or above</i>                         |                      |                      |                      |                      |                      |                      |
| SRB                                                                    | 0.005***<br>(0.001)  | 0.007***<br>(0.001)  | 0.002<br>(0.003)     | 0.003<br>(0.003)     | -0.002<br>(0.002)    | -0.005***<br>(0.002) |
| <i>Panel B: Fine and minority-share controls: High school or above</i> |                      |                      |                      |                      |                      |                      |
| SRB                                                                    | 0.005***<br>(0.001)  | 0.005***<br>(0.001)  | 0.002<br>(0.003)     | 0.003<br>(0.003)     | -0.002<br>(0.002)    | -0.005***<br>(0.002) |
| Fine                                                                   | 0.003<br>(0.004)     | 0.015***<br>(0.004)  | -0.023***<br>(0.005) | -0.004<br>(0.005)    | 0.005<br>(0.005)     | 0.003<br>(0.005)     |
| Minority share                                                         | -0.133***<br>(0.045) | -0.243***<br>(0.051) | -0.178***<br>(0.040) | -0.134***<br>(0.043) | -0.060***<br>(0.019) | -0.048**<br>(0.019)  |
| Dep. mean                                                              | 0.547                | 0.524                | 0.438                | 0.407                | 0.404                | 0.400                |
| N                                                                      | 123355               | 126417               | 96839                | 95392                | 83657                | 84248                |
| Controls                                                               | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |

Notes: SRB is standardized. Standard errors are in parentheses and clustered by individual respondent. The baseline row block reports estimates without policy controls; the second row block adds the OCP birth fine and province-cohort minority share. All regressions control for hukou status, minority status, log birth-province income per capita at age 15, and market and cohort fixed effects. Columns split by birth-province region and gender. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Overall, the robustness checks show that One-Child Policy intensity is relevant for both marriage timing and education, but it does not absorb the main SRB results. Birth fines and minority share have independent associations with the outcomes, consistent with the

Table 15: Robustness checks for education outcome: some college or above, by birth region

|                                                                         | East/Northeast      |                     | Central              |                     | West                 |                      |
|-------------------------------------------------------------------------|---------------------|---------------------|----------------------|---------------------|----------------------|----------------------|
|                                                                         | (1)                 | (2)                 | (3)                  | (4)                 | (5)                  | (6)                  |
|                                                                         | Males               | Females             | Males                | Females             | Males                | Females              |
| <i>Panel A: Baseline: Some college or above</i>                         |                     |                     |                      |                     |                      |                      |
| SRB                                                                     | 0.006***<br>(0.002) | 0.007***<br>(0.001) | 0.007***<br>(0.002)  | 0.006**<br>(0.002)  | -0.009***<br>(0.002) | -0.014***<br>(0.002) |
| <i>Panel B: Fine and minority-share controls: Some college or above</i> |                     |                     |                      |                     |                      |                      |
| SRB                                                                     | 0.007***<br>(0.002) | 0.006***<br>(0.002) | 0.007***<br>(0.002)  | 0.006***<br>(0.002) | -0.009***<br>(0.002) | -0.014***<br>(0.002) |
| Fine                                                                    | -0.009**<br>(0.004) | 0.009**<br>(0.004)  | -0.029***<br>(0.005) | 0.006<br>(0.005)    | -0.008*<br>(0.004)   | -0.011***<br>(0.004) |
| Minority share                                                          | -0.067<br>(0.046)   | -0.123**<br>(0.049) | -0.010<br>(0.036)    | -0.014<br>(0.037)   | -0.065***<br>(0.016) | -0.056***<br>(0.016) |
| Dep. mean                                                               | 0.339               | 0.342               | 0.242                | 0.234               | 0.242                | 0.249                |
| N                                                                       | 123355              | 126417              | 96839                | 95392               | 83657                | 84248                |
| Controls                                                                | Yes                 | Yes                 | Yes                  | Yes                 | Yes                  | Yes                  |

Notes: SRB is standardized. Standard errors are in parentheses and clustered by individual respondent. The baseline row block reports estimates without policy controls; the second row block adds the OCP birth fine and province-cohort minority share. All regressions control for hukou status, minority status, log birth-province income per capita at age 15, and market and cohort fixed effects. Columns split by birth-province region and gender. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

idea that fertility policy affected households through family size, resource allocation, and local institutional environments. At the same time, the SRB coefficients remain stable for age at first marriage, and the regional education patterns remain visible after adding both policy controls. These results support the interpretation that male-biased sex-ratio exposure captures marriage-market pressure beyond observable variation in fertility-policy intensity.

## 8 DiNardo-Fortin-Lemieux (DFL) Decomposition

The regression results show that sex ratio is related to marriage timing and that education delays marriage. The DFL decomposition asks how much of the change in age at first marriage can be attributed to sex-ratio changes and how much can be attributed to education upgrading. Appendix C gives the reweighting details.

Let  $Y$  denote age at first marriage,  $SR$  sex-ratio exposure, and  $Z$  education. Define

$$\mu(t_y, t_{SR}, t_Z)$$

as the mean of  $Y$  using the 1975 outcome structure, with sex-ratio and education distributions taken from the indicated cohort. The baseline mean  $\mu_0 = \mu(0, 0, 0)$  is the observed 1975 mean. The counterfactual  $\mu_1 = \mu(0, 1, 1)$  gives the 1975 cohort the 1990 sex-ratio and education distributions. The intermediate counterfactual  $\mu_2 = \mu(0, 1, 0)$  gives the 1975 cohort the 1990 sex-ratio distribution while holding education at the 1975 distribution. The total change can be written as

$$\mu_1 - \mu_0 = \underbrace{\mu_2 - \mu_0}_{\text{sex-ratio contribution}} + \underbrace{\mu_1 - \mu_2}_{\text{education contribution}}.$$

Table 16 reports the decomposition. For men, the total change is 6.65 percent. Sex-ratio exposure accounts for 45.67 percent, while education accounts for 54.33 percent. For women, the total change is 6.64 percent, with sex-ratio exposure accounting for 35.11 percent and education for 64.89 percent.

The decomposition shows that education upgrading explains a larger share of this overall delay than sex-ratio exposure. In the full sample, education accounts for 54.33 percent of the change for men and 64.89 percent for women. This is consistent with the prior results, where education is strongly associated with later marriage for both genders. Part of this relationship is mechanical: individuals who remain in school longer are less likely to marry while enrolled, so educational expansion directly shifts marriage to later ages. Education therefore matters not only as a possible response to marriage-market pressure, but also as

Table 16: DFL Decomposition of Age at First Marriage

|                            | Males |       |       | Females |       |       |
|----------------------------|-------|-------|-------|---------|-------|-------|
|                            | All   | Urban | Rural | All     | Urban | Rural |
| $\mu_0$                    | 25.04 | 26.04 | 24.15 | 23.36   | 24.35 | 22.42 |
| $\mu_1$                    | 26.71 | 27.34 | 24.51 | 24.91   | 25.39 | 22.89 |
| $\mu_2$                    | 25.80 | 26.71 | 24.24 | 23.90   | 24.68 | 22.51 |
| Total Change (%)           | 6.65  | 4.98  | 1.50  | 6.64    | 4.27  | 2.11  |
| Sex Ratio Contribution (%) | 45.67 | 51.37 | 25.08 | 35.11   | 32.21 | 18.68 |
| Education Contribution (%) | 54.33 | 48.63 | 74.92 | 64.89   | 67.79 | 81.32 |

*Notes:* DFL decomposition for the 1975 cohort using the 2010 final v2 province-urban sample.  $\mu_0$  is observed;  $\mu_1$  reweights to 1990 education and SRB;  $\mu_2$  reweights to 1990 SRB holding education at 1975. Propensity scores use HS, HS+, and standardized SRB.

an important proximate driver of delayed marriage across cohorts.

The sex-ratio contribution is smaller overall, but it is larger for men than for women. In the full sample, sex-ratio exposure accounts for 45.67 percent of the change in male age at first marriage, compared with 35.11 percent for women. This gender difference is consistent with the marriage-squeeze mechanism. Since men are on the long side of the male-biased marriage market, they face stronger competition when the sex ratio is higher.

The urban-rural split further clarifies this pattern. For urban men, the sex-ratio contribution is 51.37 percent, compared with 25.08 percent for rural men. This suggests that urban marriage timing is more responsive to demographic pressure, while rural marriage timing may be more constrained by local norms, including earlier marriage norms. In rural areas, education accounts for a much larger share of the change: 74.92 percent for men and 81.32 percent for women. This is consistent with the idea that, when age at marriage is less flexible in response to demographic pressure, education can still delay marriage through schooling itself. In urban areas, age at marriage is relatively more flexible, so demographic pressure can translate more directly into delayed marriage. The decomposition therefore reinforces the main results: education is the larger force behind the overall shift in marriage timing, but sex-ratio pressure plays a distinct and stronger role for men, especially in urban markets.

## 9 Conclusion

I study whether China’s imbalanced sex ratios shaped schooling through marriage-market incentives. Using 2010 census microdata and sex-ratio-at-birth exposure measured at the province-by-urban birth-market level, I document three findings.

First, the marriage squeeze is visible mainly in marriage timing and matching patterns. Although higher sex-ratio exposure has little effect on the probability of remaining never married, it delays age at first marriage and increases the spouse age gap for men. These results suggest that men in more male-biased markets do not primarily adjust through non-marriage in this relatively young sample. Instead, the evidence points to adjustment along the timing and matching margins: men marry later and, conditional on marriage, match with relatively younger wives. Women, in contrast, are less affected. The DFL decomposition reinforces this interpretation: sex-ratio exposure accounts for a larger share of the change in age at first marriage for men than for women, and for urban residents than for rural residents, although education upgrading accounts for a larger share of the overall shift.

Second, higher sex-ratio exposure is associated with a modest upward shift in educational attainment. A one-standard-deviation increase in SRB raises the probability of high school or above by about 0.3 percentage points for both men and women. Moving from one of the least biased to one of the most biased birth markets is roughly a four-standard-deviation change, which implies about 1.2 percentage points at the high-school threshold. The ordered-probit estimates imply a similar shift: about two percentage points of the education distribution move away from less than high school and toward higher schooling categories. Given the size of China's birth cohorts, even a one- or two-percentage-point change can correspond to potentially millions of people across the affected cohorts.

Third, the marriage-market return to education appears to operate through gender-specific channels. Both men and women may have incentives to invest in education, but the relevant returns differ across the two sides of the market. For men, who face stronger competition in male-biased markets, education appears to help on the matching margin: it is strongly associated with spouse education and partly offsets marriage delay in high-sex-ratio markets. For women, the return to education is more about match quality and household bargaining position. Education strongly predicts spouse education, consistent with positive assortative mating, and the household-headship results suggest that education is positively associated with women's intra-household status, although headship is only a proxy for bargaining position.

Taken together, the results suggest that demographic imbalance can affect human-capital investment through the marriage market, but the education response is not uniform across places. The estimates do not imply that marriage-market pressure was the dominant driver of China's educational expansion as the estimates are small. Local norms, economic development, education supply, and household resources remain central determinants of schooling. Rather, the results show that male-biased marriage markets created an additional incentive to invest in education, especially in markets where schooling was accessible and valued in

the marriage market. This interpretation is supported by the heterogeneity and robustness checks: the education response is positive in urban markets and in East/Northeast and Central provinces, weaker or negative in rural and Western markets, and the main regional patterns are preserved after controlling for One-Child Policy intensity and province-cohort minority share.

Overall, I show that skewed sex ratios affect more than marriage formation. By using sex ratios at birth to measure cohort-level marriage-market tightness, I link China's demographic imbalance to both marriage outcomes and human-capital investment. The results provide a complementary explanation for educational upgrading in China: alongside education policy, economic development, and quantity-quality responses to fertility decline, the marriage squeeze created an additional incentive to invest in schooling. I also show that the marriage-market return to education is gender-specific. Hence, the results add to the understanding of the long-run consequences of the One-Child Policy and shed light on how demographic shocks can reshape human-capital investment through the marriage market.

# A Data Sources, Preprocessing, and Sample Construction

## A.1 Data sources

I use sampled census microdata from the 1990, 2000, and 2010 population censuses.<sup>10</sup> The sex-ratio measures are constructed from the 1990 and 2000 census microdata samples. The main individual-level analysis uses the 2010 census microdata sample. These files contain individual-level information on demographic characteristics, education, marital status, hukou status, birthplace, and current residence.

For the 1990 census, I use the 1 percent census microdata sample. The IPUMS metadata report 11,835,947 person records ([National Bureau of Statistics of China and IPUMS International, 1990](#)). For the 2000 census, I use the 1 percent census microdata sample from the Fifth Population Census, with 11,804,344 records reported by CNDeepData ([CNDeepData, 2023](#)). For the 2010 census, I use the 1.58 percent systematic sample of long-form records from the Sixth Population Census, which contains 2,105,260 observations ([CNDeepData, 2023](#)). Long-form respondents answered a more detailed questionnaire, including questions on demographic characteristics, work, education, and household information. The 2010 census long form was administered to 10 percent of households. After restricting the analysis sample to cohorts born between 1975 and 1990, the sample size is 612,787 observations.

The data do not include sampling weights. I therefore assign equal weight to each individual in the analysis, following guidance I received from staff at the National Bureau of Statistics.

## A.2 Regional classification and background statistics

The regional heterogeneity analysis uses the National Bureau of Statistics’ four-region economic classification. The eastern region includes Beijing, Tianjin, Hebei, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, and Hainan. The central region includes Shanxi, Anhui, Jiangxi, Henan, Hubei, and Hunan. The western region includes Inner Mongolia, Guangxi, Chongqing, Sichuan, Guizhou, Yunnan, Tibet, Shaanxi, Gansu, Qinghai, Ningxia, and Xinjiang. The northeastern region includes Liaoning, Jilin, and Heilongjiang.

---

<sup>10</sup>The 1990 file is based on the IPUMS China 1990 Fourth National Population Census 1 percent sample, originally produced by the National Bureau of Statistics ([National Bureau of Statistics of China and IPUMS International, 1990](#)); the 1990 cleaning follows the procedure described in a Jingguan Zhijia forum post on the Stata/SPSS version of the IPUMS file ([Jingguan Zhijia, 2021](#)). The 2000 and 2010 files are obtained from CNDeepData’s population census microdata collection, which reports the National Bureau of Statistics as the original data source ([CNDeepData, 2023](#)).

Table 17 summarizes the regional background statistics used to motivate the birth-region heterogeneity analysis. The four regions differ sharply in economic development, urbanization, and schooling conditions. The eastern region has the highest per-capita GDP and household income. The Northeast has lower household income than the East but remains highly urbanized and has a population-adjusted college-plus attainment level close to the East. Central and Western China have lower urbanization and lower junior-college-or-above attainment per capita.

Table 17: Regional background statistics, 2010

| Region       | Per-capita GDP<br>yuan | Urban income<br>yuan | Rural income<br>yuan | Urbanization<br>percent | Junior college+<br>per 100,000 |
|--------------|------------------------|----------------------|----------------------|-------------------------|--------------------------------|
| Eastern      | 46,354                 | 23,273               | 8,143                | 59.8                    | 10,465                         |
| Central      | 24,242                 | 15,962               | 5,510                | 43.6                    | 7,460                          |
| Western      | 22,476                 | 15,806               | 4,418                | 41.4                    | 7,503                          |
| Northeastern | 34,303                 | 15,941               | 6,434                | 57.7                    | 10,431                         |

Notes: Per-capita GDP, urban household disposable income, and rural net income are from National Bureau of Statistics, *China Statistical Yearbook 2011*, Table 1-6, “Main Indicators of National Economic and Social Development by Eastern, Central, Western and Northeastern Provinces (2010).” Urbanization rates are population-weighted regional averages computed from province-level urbanization rates in *China Statistical Yearbook 2013*, Table 3-6, using 2010 province populations from *China Statistical Yearbook 2011*, Table 3-4. Junior-college-and-above attainment is a population-weighted regional average computed from *China Statistical Yearbook 2011*, Table 3-10, using the same population weights.

I combine the eastern and northeastern regions in the regression because the relevant empirical contrast is schooling access and urban opportunity sets, not identical economic history. The Northeast is an old industrial region with a different development path and weaker household-income performance than the eastern coast. However, in 2010 it was close to the East in urbanization and junior-college-or-above attainment per capita. I therefore interpret the combined East/Northeast category as a high-urbanization and high-schooling-infrastructure group, while treating Central and Western China as less urbanized and lower-schooling-access comparison regions.

### A.3 Harmonizing province boundaries

I harmonize province definitions before constructing sex-ratio measures. The main adjustment concerns Chongqing, which became a centrally administered municipality in 1997. Because Chongqing was part of Sichuan in the 1990 census but appears as a separate province-level unit in later data, I use address codes to reconstruct a consistent Chongqing definition.

I reconstruct the later Chongqing area in the 1990 data from the relevant Sichuan prefectures. In the 1990 data, the harmonized Chongqing definition includes original Sichuan



records with prefecture codes for Chongqing city, Wanxian, Fuling, and Qianjiang. Equivalently, in the six-digit address-code system this corresponds to the following address-code ranges:

Table 18: Address-code components used to reconstruct Chongqing

| Address-code range | Province code | Prefecture code | Corresponding area                     |
|--------------------|---------------|-----------------|----------------------------------------|
| 510201–510232      | 51            | 02              | Chongqing city and affiliated counties |
| 512201–512229      | 51            | 22              | Wanxian city/prefecture                |
| 512301–512326      | 51            | 23              | Fuling city/prefecture                 |
| 513521–513525      | 51            | 35              | Qianjiang prefecture                   |

Notes: These codes describe the harmonized 1990-style Chongqing construction used to align pre-1997 Sichuan records with the post-1997 Chongqing municipality.

## A.4 Urban/rural classification

I classify cities and towns as urban and villages as rural, following the National Bureau of Statistics urban-rural classification rules. The operational coding is census-specific because the available variables differ across census years. The 1990 and 2000 censuses include variables that directly indicate urban or rural residence. The 2010 census does not include an equivalent variable in the microdata I use, so I infer urban or rural residence from digits 10–12 of the address code. Codes 001–199 and 400–499 are classified as urban, while codes 200–399 and 500–599 are classified as rural ([National Bureau of Statistics of China, 2009](#)).

It is important to note that the official urban/rural classification changed across the census years I use. According to the National Bureau of Statistics, the 1990 census period followed the earlier classification framework introduced in 1955, which relied mainly on population size and the share of non-agricultural population, with residential settlements as the basic classification unit ([National Bureau of Statistics of China, 2025](#)). For the Fifth Population Census in 2000, the National Bureau of Statistics introduced the 1999 trial regulation on statistical urban/rural classification ([National Bureau of Statistics of China, 1999, 2025](#)). This regulation classified areas into urban and rural categories, with urban areas including cities and towns and rural areas defined as areas outside the classified urban category. For the 2010 census period, the 2008 regulation provided the formal classification basis ([National Bureau of Statistics of China, 2008](#)). Compared with the earlier system, the 2008 regulation defined urban and rural areas using administrative divisions, residents’ and villagers’ committee jurisdictions, and the physical connection of built-up areas. Under this

rule, urban areas include city districts and town districts, while rural areas are areas outside the officially classified urban area.

I calculate the urban share in each census year using these census-specific definitions. The resulting urban share is substantially higher in 2010 than in 1990 and 2000. This increase reflects both actual urbanization over the sample period and changes in the statistical definition of urban and rural areas across censuses. This matters for my sex-ratio assignment because the 2010 urban/rural classification is used to infer respondents’ birth-market type. If places or individuals classified as rural in 1990 or 2000 are classified as urban by 2010, some respondents may be assigned the urban sex ratio even though their childhood environment was rural. This measurement issue may overstate exposure to urban birth markets and attenuate or otherwise change the interpretation of the estimated urban-rural heterogeneity. I discuss this more in the following section.

Table 19: Urban/rural shares under selected census-specific definitions

| Year | Sampling rate                                | Urban share | Rural share | Observations |
|------|----------------------------------------------|-------------|-------------|--------------|
| 1990 | 1% census sample                             | 35.84       | 64.16       | 11,835,947   |
| 2000 | 1% census sample                             | 36.50       | 63.50       | 11,804,344   |
| 2010 | 1.58% systematic sample of long-form records | 44.12       | 55.88       | 2,105,260    |

Notes: Sampling-rate sources are IPUMS metadata for 1990 and CNDeepData data descriptions for 2000 and 2010 ([National Bureau of Statistics of China and IPUMS International, 1990](#); [CNDeepData, 2023](#)). For 2010, the listed rate refers to CNDeepData’s systematic sample of long-form records; CNDeepData reports that the census long form was administered to 10 percent of households.

## A.5 Constructing and assigning sex-ratio exposure

The main exposure variable is the sex ratio at birth, measured at the province-by-urban/rural cohort-market level. For each birth cohort, I compute the male-to-female ratio using a seven-year rolling window centered on the cohort year. The window includes cohorts born within three years of the respondent’s birth year. The province-by-urban/rural SRB panel is constructed from the 1990 and 2000 census microdata samples.

Since the 2010 census reports respondents’ province of birth, I can assign sex-ratio exposure by birth province. However, I cannot observe whether respondents were born in urban or rural areas. I therefore combine birth province with respondents’ current urban/rural status to construct an inferred birth-market measure. This assignment keeps the treatment predetermined by birth province and cohort, but it uses current residence to proxy for the missing childhood urban/rural market.

This limitation is more important for education than for age at marriage. Age at marriage is realized in adulthood, when current residence may be closer to the relevant marriage-market location. Educational investments, however, are more likely to be shaped by childhood location, schooling access, and household environment. If rural-born individuals later move to urban areas, the current-residence assignment may classify them as facing an urban birth market and therefore overstate the number of respondents assigned to urban birth markets.

## A.6 Rural-urban migration and hukou-based robustness

The main province-by-urban/rural assignment implicitly assumes that current urban/rural residence is informative about the respondent’s childhood market. This assumption may be violated by rural-urban migration and urbanization over the 1975–1990 birth cohorts. The likely direction of the concern is that the main assignment may overstate urban birth-market exposure, because some respondents currently living in urban areas were born and educated in rural areas.

The scale and composition of migration in 2010 suggest that this measurement concern is quantitatively relevant. [Zheng and Yang \(2013\)](#) report that China’s floating population, defined by those leaving hukou registration place for more than six months, increased from 6.57 million in 1982 to about 220 million in 2010, or roughly 17 percent of the national population. They also report that about 150 million members of the 2010 floating population were from rural areas, and that rural-origin individuals accounted for 54 percent of within-province migrants and nearly 82 percent of cross-province migrants. These figures indicate that a large share of migrants observed around the 2010 census were rural-origin individuals, which is the group most likely to generate misclassification when current urban/rural residence is used to proxy for childhood urban/rural status. [Wu et al. \(2022\)](#) provide a more direct estimate of rural-to-urban population transfer. Using census and statistical data, they estimate that rural-to-urban population transfer increased China’s urban population by 90.72 million during 2005–2010 and by 79.40 million during 2010–2015. Rural-to-urban mobility is more likely to classify rural-origin respondents as urban in the main assignment, so the current-residence measure may overstate exposure to urban birth markets.

To address this concern, I also construct a hukou-based assignment as a robustness exercise. Agricultural hukou is treated as rural and non-agricultural hukou as urban. Because hukou status is assigned at birth based on parents’ hukou status, it may be more informative about origin status than current residence. This alternative is also imperfect because hukou status can change. However, non-agricultural hukou conversion remained restricted

around 2010, so hukou type provides a useful proxy for urban/rural origin. The resulting hukou-based province-by-urban/rural SRB regressions are consistent with the main results. I nevertheless treat this exercise as a robustness check rather than as the preferred measure. Hukou status is itself tied to education access and other institutional advantages, so I prefer to include hukou as a control in the main regressions rather than use it to define the main SRB assignment.

## A.7 Cross-province migration

In addition to rural-urban migration, cross-province migration can also affect the interpretation of the sex-ratio assignment. The main exposure is assigned using birth province, which is the relevant predetermined location observed in the census. However, adult outcomes and current urban/rural residence may partly reflect destination markets if respondents moved across provinces. The ideal data would identify each respondent’s birth location, childhood urban/rural status, full migration history, and the timing of migration relative to schooling. In particular, it would be important to know whether migration occurred before, during, or after schooling was completed. Migration after schooling would make current residence partly an outcome of education, while migration before or during schooling could expose individuals to destination schooling opportunities and destination marriage-market expectations.

The 2010 census does not provide the full migration path or the timing of migration relative to schooling. It reports birth province, current province, hukou status, current urban/rural status, and whether the respondent left the hukou registration place for at least six months. I therefore use these variables to describe the scope of the concern rather than to fully correct for migration selection.

In the main regression sample, 94,908 respondents, or 15.52 percent of the analysis sample, live outside their birth province. Table 20 compares the four-category completed-education distribution by gender and cross-province migration status. Cross-province migrants have higher BA+ shares than birth-province stayers: 11.82 percent versus 7.90 percent among men, and 10.85 percent versus 7.98 percent among women. The broader distribution is nevertheless similar across groups. For example, less-than-high-school shares are 50.09 percent for male migrants and 53.29 percent for male stayers, and 53.13 percent for female migrants and 55.01 percent for female stayers.

These patterns suggest that cross-province migration can affect the interpretation of the estimates, especially if migration is selected on education or if migration occurred before schooling decisions were completed. At the same time, most respondents remain in their birth

Table 20: Completed education by cross-province migration status

|              | Stayer  |         | Cross-province migrant |        |
|--------------|---------|---------|------------------------|--------|
|              | Male    | Female  | Male                   | Female |
| Less than HS | 53.29   | 55.01   | 50.09                  | 53.13  |
| HS           | 18.95   | 16.85   | 20.33                  | 18.33  |
| Some College | 19.86   | 20.16   | 17.76                  | 17.69  |
| BA+          | 7.90    | 7.98    | 11.82                  | 10.85  |
| N            | 254,507 | 261,929 | 50,125                 | 44,783 |

Notes: The sample is the 2010 main regression sample, restricted to cohorts 1975–1990. Stayers have the same birth province and current province; cross-province migrants have different birth and current provinces.

province, and the education distribution of migrants and stayers is not sharply different in the main regression sample. I therefore interpret the province-by-urban/rural SRB as an inferred birth-market exposure. The descriptive migration statistics and heterogeneity exercises help assess the scope of the concern, but they should not be interpreted as a full correction for migration selection because the census does not identify the timing of migration relative to schooling. When restricting the sample to birth-province stayers, the main results are robust and remain statistically significant.

## A.8 Policy fines

The policy-fine data come from the replication files for [Ebenstein \(2010\)](#), downloaded from Ebenstein’s webpage. It is a province-by-year panel for 31 province-level units and birth years 1979–2000.

The baseline policy file contains `fine`, `premium`, `bonus`, and `policy`. In Ebenstein’s code, `fine` is labelled as the fine in years of income; `premium` indicates whether higher-order births face a premium fine regime; `bonus` records the one-child reward or bonus amount; and `policy` records the provincial fertility-policy rule. The policy-rule variable takes the values 1, 1.5, and 2, corresponding to one-child, one-and-a-half-child, and two-child/minority-policy regimes. The code fills missing fine and bonus values forward within province when no new report is available. The accompanying birth-fine program then uses the province-year fine schedule and policy rules to assign fines by birth order, province, birth year, family type, urban/rural hukou, and parity. The program notes state that the fine is measured in years of household income.

For my analysis, I merge the province-year policy panel to the 2010 census sample by

respondents' birth province and birth year. Because the Ebenstein panel begins in 1979, cohorts born before 1979 are assigned zero fine exposure. This coding is relevant for the 1975–1990 cohorts used in the main 2010 analysis: cohorts 1975–1978 have zero fines, while cohorts 1979 onward are matched to the province-year fine schedule.

Following the robustness specifications in this paper, I use only **fine**, the one-child-policy birth fine measured in years of household income, as the policy-intensity control. The other variables in the replication file, including the premium-fine indicator, the one-child bonus amount, and policy-type indicators, are retained in the merged data for transparency but are not included in the main robustness specifications.

## A.9 Local income at age 15

The baseline controls include log local income when the respondent was age 15. This variable is intended to absorb local economic conditions during adolescence, around the period when high-school continuation and postsecondary-entry decisions become relevant. The source data are province-by-year-by-urban/rural income and consumption series downloaded from National Bureau of Statistics webpages.

I first construct a province-year-urban/rural panel covering 31 provinces, years 1978–2010, and two urban/rural cells. Nominal per-capita income is converted to 2000 RMB using the resident CPI series, with CPI normalized so that the 2000 value equals 434.0. The conversion is

$$\text{income}_{p,u,t}^{2000} = \frac{\text{income}_{p,u,t}^{\text{nominal}}}{\text{CPI}_t} \times 434.0,$$

where  $p$  indexes province,  $u$  indexes urban/rural status, and  $t$  indexes calendar year. I then take logs to obtain  $\log(\text{income}_{p,u,t}^{2000})$ .

For each respondent  $i$ , I assign the income control using the birth province, the respondent's current urban/rural status, and the calendar year when the respondent was age 15:

$$t_i^{15} = \text{birth year}_i + 15.$$

The resulting control is

$$\text{Log income at age 15}_i = \log(\text{income}_{p_i^{\text{birth}}, u_i^{\text{current}}, t_i^{15}}^{2000}).$$

Thus, for a respondent born in 1985, the age-15 income control is measured in 2000 and assigned from the respondent's birth province and current urban/rural status.

This control shares the same urban/rural limitation as the SRB assignment: because

urban/rural status at birth is not observed, the income cell uses current urban/rural status as the inferred childhood market.

## B Appendix Figures and Tables

### B.1 Education by schooling-completion status

Table 21: Some college or above by schooling status

|            | Completed school     |                      | In school            |                     | Incomplete/dropout  |                     |
|------------|----------------------|----------------------|----------------------|---------------------|---------------------|---------------------|
|            | (1)                  | (2)                  | (3)                  | (4)                 | (5)                 | (6)                 |
|            | Males                | Females              | Males                | Females             | Males               | Females             |
| SRB        | 0.005***<br>(0.001)  | 0.003**<br>(0.001)   | -0.001<br>(0.002)    | -0.005**<br>(0.002) | 0.009*<br>(0.006)   | 0.000<br>(0.005)    |
| Hukou      | 0.388***<br>(0.002)  | 0.385***<br>(0.002)  | 0.095***<br>(0.004)  | 0.095***<br>(0.004) | 0.161***<br>(0.019) | 0.135***<br>(0.019) |
| Minority   | -0.008***<br>(0.003) | -0.006*<br>(0.003)   | 0.000<br>(0.008)     | 0.005<br>(0.007)    | -0.017**<br>(0.007) | -0.008<br>(0.007)   |
| Log income | -0.052***<br>(0.009) | 0.044***<br>(0.009)  | 0.303***<br>(0.056)  | 0.205***<br>(0.056) | 0.045<br>(0.038)    | 0.044<br>(0.033)    |
| Constant   | 0.507***<br>(0.072)  | -0.263***<br>(0.072) | -1.802***<br>(0.480) | -0.942**<br>(0.481) | -0.327<br>(0.297)   | -0.320<br>(0.254)   |
| Controls   | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 |
| Dep. mean  | 0.210                | 0.215                | 0.870                | 0.884               | 0.040               | 0.029               |
| N          | 263,051              | 266,408              | 34,231               | 32,788              | 4,861               | 4,561               |
| R-squared  | 0.299                | 0.314                | 0.103                | 0.101               | 0.124               | 0.117               |

Notes: SRB is standardized. The dependent variable is Some college or above. Completed school restricts to completed\_school==1; in school restricts to school\_status==1; incomplete/dropout combines school\_status==3 and school\_status==4. All regressions include controls as well as market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

### B.2 Full heterogeneity estimates



Table 22: Exact some-college category among in-school and non-completion samples

|            | In school             |                      | Incomplete or dropped out |                     |
|------------|-----------------------|----------------------|---------------------------|---------------------|
|            | (1)<br>Males          | (2)<br>Females       | (3)<br>Males              | (4)<br>Females      |
| SRB        | 0.005<br>(0.004)      | 0.005<br>(0.004)     | 0.009*<br>(0.006)         | -0.000<br>(0.005)   |
| Hukou      | 0.053***<br>(0.004)   | 0.046***<br>(0.004)  | 0.158***<br>(0.019)       | 0.122***<br>(0.019) |
| Minority   | 0.010<br>(0.009)      | -0.007<br>(0.009)    | -0.017**<br>(0.007)       | -0.008<br>(0.007)   |
| Log income | 1.418***<br>(0.075)   | 1.170***<br>(0.081)  | 0.049<br>(0.038)          | 0.048<br>(0.032)    |
| Constant   | -11.479***<br>(0.651) | -9.331***<br>(0.699) | -0.360<br>(0.296)         | -0.351<br>(0.249)   |
| Controls   | Yes                   | Yes                  | Yes                       | Yes                 |
| Dep. mean  | 0.815                 | 0.826                | 0.039                     | 0.028               |
| N          | 34,231                | 32,788               | 4,861                     | 4,561               |
| R-squared  | 0.124                 | 0.132                | 0.120                     | 0.112               |

Notes: SRB is standardized. The dependent variable equals one for the exact Some college category. In school restricts to school\_status==1. Incomplete or dropped out combines school\_status==3 (incomplete schooling) and school\_status==4 (dropped out). Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 23: Full heterogeneity estimates by birth-province region

*Panel A: High school or above*

|            | East/Northeast      |                     | Central             |                     | West                 |                      |
|------------|---------------------|---------------------|---------------------|---------------------|----------------------|----------------------|
|            | (1)                 | (2)                 | (3)                 | (4)                 | (5)                  | (6)                  |
|            | Males               | Females             | Males               | Females             | Males                | Females              |
| SRB        | 0.005***<br>(0.001) | 0.007***<br>(0.001) | 0.002<br>(0.003)    | 0.003<br>(0.003)    | -0.002<br>(0.002)    | -0.005***<br>(0.002) |
| Hukou      | 0.390***<br>(0.003) | 0.410***<br>(0.003) | 0.436***<br>(0.004) | 0.443***<br>(0.004) | 0.440***<br>(0.004)  | 0.459***<br>(0.004)  |
| Minority   | -0.014**<br>(0.007) | -0.008<br>(0.007)   | -0.015<br>(0.012)   | -0.013<br>(0.011)   | -0.031***<br>(0.004) | -0.034***<br>(0.004) |
| Log income | 0.041***<br>(0.014) | 0.061***<br>(0.014) | -0.037*<br>(0.022)  | -0.016<br>(0.021)   | 0.036*<br>(0.020)    | 0.110***<br>(0.019)  |
| Constant   | 0.051<br>(0.119)    | -0.148<br>(0.114)   | 0.613***<br>(0.177) | 0.405**<br>(0.170)  | -0.001<br>(0.157)    | -0.610***<br>(0.151) |
| Dep. mean  | 0.547               | 0.524               | 0.438               | 0.407               | 0.404                | 0.400                |
| N          | 123,355             | 126,417             | 96,839              | 95,392              | 83,657               | 84,248               |
| Controls   | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  | Yes                  |

*Panel B: Some college or above*

|            | East/Northeast       |                      | Central              |                      | West                 |                      |
|------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|            | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|            | Males                | Females              | Males                | Females              | Males                | Females              |
| SRB        | 0.006***<br>(0.002)  | 0.007***<br>(0.001)  | 0.007***<br>(0.002)  | 0.006**<br>(0.002)   | -0.009***<br>(0.002) | -0.014***<br>(0.002) |
| Hukou      | 0.390***<br>(0.003)  | 0.392***<br>(0.003)  | 0.411***<br>(0.004)  | 0.400***<br>(0.004)  | 0.412***<br>(0.004)  | 0.421***<br>(0.004)  |
| Minority   | 0.006<br>(0.007)     | 0.027***<br>(0.007)  | -0.002<br>(0.010)    | -0.001<br>(0.010)    | -0.004<br>(0.004)    | -0.009**<br>(0.004)  |
| Log income | 0.159***<br>(0.014)  | 0.235***<br>(0.013)  | 0.100***<br>(0.020)  | 0.150***<br>(0.020)  | 0.152***<br>(0.018)  | 0.245***<br>(0.018)  |
| Constant   | -1.144***<br>(0.115) | -1.774***<br>(0.111) | -0.665***<br>(0.157) | -1.082***<br>(0.155) | -1.074***<br>(0.141) | -1.819***<br>(0.139) |
| Dep. mean  | 0.339                | 0.342                | 0.242                | 0.234                | 0.242                | 0.249                |
| N          | 123,355              | 126,417              | 96,839               | 95,392               | 83,657               | 84,248               |
| Controls   | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |

Notes: SRB is standardized. Columns split by birth-province region and gender. East/Northeast combines Northeast with East. All regressions include controls as well as market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 24: Full heterogeneity estimates by urban status

*Panel A: High school or above*

|            | Urban                |                      | Rural                |                      |
|------------|----------------------|----------------------|----------------------|----------------------|
|            | (1)<br>Males         | (2)<br>Females       | (3)<br>Males         | (4)<br>Females       |
| SRB        | 0.005***<br>(0.001)  | 0.006***<br>(0.001)  | -0.004<br>(0.002)    | -0.007***<br>(0.002) |
| Hukou      | 0.397***<br>(0.002)  | 0.416***<br>(0.002)  | 0.469***<br>(0.004)  | 0.479***<br>(0.004)  |
| Minority   | -0.020***<br>(0.006) | -0.025***<br>(0.005) | -0.030***<br>(0.004) | -0.026***<br>(0.004) |
| Log income | 0.069***<br>(0.019)  | 0.072***<br>(0.019)  | 0.142***<br>(0.015)  | 0.175***<br>(0.014)  |
| Constant   | -0.142<br>(0.167)    | -0.195<br>(0.164)    | -0.844***<br>(0.117) | -1.128***<br>(0.110) |
| Dep. mean  | 0.696                | 0.687                | 0.275                | 0.239                |
| N          | 143,135              | 146,413              | 160,716              | 159,644              |
| Controls   | Yes                  | Yes                  | Yes                  | Yes                  |

*Panel B: Some college or above*

|            | Urban                |                      | Rural                |                      |
|------------|----------------------|----------------------|----------------------|----------------------|
|            | (1)<br>Males         | (2)<br>Females       | (3)<br>Males         | (4)<br>Females       |
| SRB        | 0.005***<br>(0.001)  | 0.005***<br>(0.001)  | -0.009***<br>(0.002) | -0.011***<br>(0.002) |
| Hukou      | 0.407***<br>(0.002)  | 0.412***<br>(0.002)  | 0.400***<br>(0.004)  | 0.393***<br>(0.004)  |
| Minority   | 0.006<br>(0.006)     | 0.005<br>(0.006)     | -0.008***<br>(0.003) | -0.005*<br>(0.003)   |
| Log income | 0.121***<br>(0.021)  | 0.198***<br>(0.021)  | 0.084***<br>(0.011)  | 0.114***<br>(0.010)  |
| Constant   | -0.831***<br>(0.182) | -1.503***<br>(0.177) | -0.551***<br>(0.081) | -0.782***<br>(0.080) |
| Dep. mean  | 0.459                | 0.464                | 0.123                | 0.116                |
| N          | 143,135              | 146,413              | 160,716              | 159,644              |
| Controls   | Yes                  | Yes                  | Yes                  | Yes                  |

Notes: SRB is standardized. Columns split by urban status and gender. All regressions include controls as well as market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 25: Full heterogeneity estimates by migrant type

Panel A: High school or above

|            | Type 1               |                      | Type 2               |                      | Type 3               |                     | Type 4              |                     |
|------------|----------------------|----------------------|----------------------|----------------------|----------------------|---------------------|---------------------|---------------------|
|            | (1)<br>Males         | (2)<br>Females       | (3)<br>Males         | (4)<br>Females       | (5)<br>Males         | (6)<br>Females      | (7)<br>Males        | (8)<br>Females      |
| SRB        | 0.006***<br>(0.002)  | 0.006***<br>(0.002)  | 0.002<br>(0.002)     | 0.003*<br>(0.002)    | 0.001<br>(0.003)     | -0.002<br>(0.003)   | 0.009**<br>(0.004)  | 0.008<br>(0.005)    |
| Hukou      | 0.417***<br>(0.004)  | 0.432***<br>(0.004)  | 0.384***<br>(0.003)  | 0.413***<br>(0.003)  | 0.466***<br>(0.005)  | 0.473***<br>(0.005) | 0.508***<br>(0.008) | 0.505***<br>(0.009) |
| Minority   | -0.019***<br>(0.005) | -0.017***<br>(0.005) | -0.030***<br>(0.006) | -0.041***<br>(0.005) | -0.028***<br>(0.010) | -0.019*<br>(0.010)  | -0.035**<br>(0.017) | -0.011<br>(0.015)   |
| Log income | 0.032**<br>(0.013)   | 0.068***<br>(0.013)  | 0.011<br>(0.017)     | 0.065***<br>(0.016)  | 0.057*<br>(0.033)    | 0.026<br>(0.033)    | -0.109*<br>(0.059)  | 0.070<br>(0.055)    |
| Constant   | 0.024<br>(0.103)     | -0.284***<br>(0.102) | 0.317**<br>(0.141)   | -0.166<br>(0.129)    | -0.116<br>(0.270)    | 0.099<br>(0.269)    | 1.303***<br>(0.499) | -0.255<br>(0.459)   |
| Dep. mean  | 0.413                | 0.401                | 0.546                | 0.512                | 0.465                | 0.439               | 0.622               | 0.568               |
| N          | 148,451              | 144,550              | 105,415              | 116,855              | 38,911               | 34,064              | 11,072              | 10,588              |
| Controls   | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 |

Panel B: Some college or above

|            | Type 1               |                      | Type 2               |                      | Type 3              |                      | Type 4              |                      |
|------------|----------------------|----------------------|----------------------|----------------------|---------------------|----------------------|---------------------|----------------------|
|            | (1)<br>Males         | (2)<br>Females       | (3)<br>Males         | (4)<br>Females       | (5)<br>Males        | (6)<br>Females       | (7)<br>Males        | (8)<br>Females       |
| SRB        | 0.003**<br>(0.002)   | 0.002<br>(0.002)     | 0.004**<br>(0.002)   | 0.001<br>(0.002)     | -0.003<br>(0.003)   | -0.005*<br>(0.003)   | -0.001<br>(0.004)   | 0.001<br>(0.005)     |
| Hukou      | 0.341***<br>(0.004)  | 0.356***<br>(0.004)  | 0.379***<br>(0.003)  | 0.385***<br>(0.003)  | 0.524***<br>(0.006) | 0.495***<br>(0.006)  | 0.635***<br>(0.008) | 0.593***<br>(0.009)  |
| Minority   | -0.003<br>(0.004)    | 0.011**<br>(0.004)   | 0.000<br>(0.006)     | -0.016***<br>(0.005) | -0.007<br>(0.008)   | 0.010<br>(0.008)     | 0.015<br>(0.015)    | 0.017<br>(0.013)     |
| Log income | 0.154***<br>(0.012)  | 0.243***<br>(0.012)  | 0.154***<br>(0.016)  | 0.226***<br>(0.015)  | 0.070***<br>(0.026) | 0.093***<br>(0.028)  | 0.009<br>(0.054)    | 0.239***<br>(0.052)  |
| Constant   | -1.109***<br>(0.097) | -1.815***<br>(0.098) | -1.047***<br>(0.134) | -1.661***<br>(0.123) | -0.451**<br>(0.216) | -0.651***<br>(0.229) | 0.076<br>(0.456)    | -1.869***<br>(0.429) |
| Dep. mean  | 0.229                | 0.241                | 0.348                | 0.333                | 0.254               | 0.249                | 0.447               | 0.407                |
| N          | 148,451              | 144,550              | 105,415              | 116,855              | 38,911              | 34,064               | 11,072              | 10,588               |
| Controls   | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                  | Yes                 | Yes                  |

Notes: SRB is standardized. Columns split by migrant type and gender. Type 1: birth-province stayers; Type 2: within-province temporary migration; Type 3: cross-province temporary migration; Type 4: permanent cross-province migration. All regressions include controls as well as market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 26: Full heterogeneity estimates by birth-province population size

*Panel A: High school or above*

|            | Large population     |                      | Small population     |                      |
|------------|----------------------|----------------------|----------------------|----------------------|
|            | (1)<br>Males         | (2)<br>Females       | (3)<br>Males         | (4)<br>Females       |
| SRB        | -0.001<br>(0.002)    | -0.002<br>(0.002)    | 0.007***<br>(0.001)  | 0.009***<br>(0.001)  |
| Hukou      | 0.408***<br>(0.003)  | 0.421***<br>(0.003)  | 0.431***<br>(0.003)  | 0.450***<br>(0.003)  |
| Minority   | -0.035***<br>(0.009) | -0.023***<br>(0.008) | -0.023***<br>(0.004) | -0.026***<br>(0.004) |
| Log income | 0.044**<br>(0.017)   | 0.054***<br>(0.017)  | 0.035***<br>(0.013)  | 0.087***<br>(0.012)  |
| Constant   | -0.001<br>(0.140)    | -0.119<br>(0.135)    | 0.030<br>(0.103)     | -0.409***<br>(0.099) |
| Dep. mean  | 0.475                | 0.445                | 0.471                | 0.463                |
| N          | 167,087              | 167,269              | 136,764              | 138,788              |
| Controls   | Yes                  | Yes                  | Yes                  | Yes                  |

*Panel B: Some college or above*

|            | Large population     |                      | Small population     |                      |
|------------|----------------------|----------------------|----------------------|----------------------|
|            | (1)<br>Males         | (2)<br>Females       | (3)<br>Males         | (4)<br>Females       |
| SRB        | 0.007***<br>(0.002)  | 0.008***<br>(0.002)  | 0.002<br>(0.002)     | -0.001<br>(0.001)    |
| Hukou      | 0.408***<br>(0.003)  | 0.395***<br>(0.003)  | 0.395***<br>(0.003)  | 0.412***<br>(0.003)  |
| Minority   | -0.007<br>(0.007)    | 0.006<br>(0.007)     | -0.001<br>(0.003)    | -0.001<br>(0.003)    |
| Log income | 0.094***<br>(0.015)  | 0.159***<br>(0.015)  | 0.184***<br>(0.012)  | 0.265***<br>(0.012)  |
| Constant   | -0.617***<br>(0.120) | -1.154***<br>(0.119) | -1.333***<br>(0.099) | -1.990***<br>(0.096) |
| Dep. mean  | 0.267                | 0.262                | 0.298                | 0.307                |
| N          | 167,087              | 167,269              | 136,764              | 138,788              |
| Controls   | Yes                  | Yes                  | Yes                  | Yes                  |

Notes: SRB is standardized. Columns split by birth-province population size and gender. Large/small split at median birth-province population. All regressions include controls as well as market and cohort fixed effects. Standard errors are in parentheses and clustered by individual respondent. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## C DFL decomposition details

This appendix provides the reweighting details for the DiNardo-Fortin-Lemieux decomposition reported in Section 8. The purpose is to quantify how much of the cohort change in age at first marriage is associated with changes in sex-ratio exposure and how much is associated with educational upgrading.

Consider two cohorts, indexed by  $t \in \{0, 1\}$ , where  $t = 0$  denotes the 1975 cohort and  $t = 1$  denotes the 1990 cohort. Let  $Y$  denote age at first marriage. Let  $X = (SR, Z)$ , where  $SR$  is standardized sex-ratio exposure and  $Z$  contains the education variables used in the decomposition. The observed density of  $Y$  in cohort  $t$  can be written as

$$f_t(y) = \int f_t(y | x) dF_t(x),$$

where  $f_t(y | x)$  is the conditional outcome distribution and  $F_t(x)$  is the cohort-specific distribution of covariates.

The DFL counterfactual asks what the mean age at first marriage would be for the 1975 cohort if it faced the covariate distribution of the 1990 cohort while preserving the 1975 conditional outcome structure. The full counterfactual density is

$$f^{CF}(y) = \int f_0(y | x) dF_1(x).$$

Using reweighting, this can be written as

$$f^{CF}(y) = \int f_0(y | x) \omega(x) dF_0(x),$$

where the full reweighting factor is

$$\omega(x) = \frac{dF_1(x)}{dF_0(x)}.$$

Following the DFL approach, I estimate this weight using a logit model for cohort membership. Let  $p(x) = \Pr(t = 1 | x)$ . By Bayes' rule,

$$\omega(x) = \frac{p(x)}{1 - p(x)} \frac{\Pr(t = 0)}{\Pr(t = 1)}.$$

Observations in the 1975 cohort that resemble the 1990 cohort in terms of observed characteristics therefore receive larger weights.

To isolate the contribution of sex-ratio exposure, partition the covariates into  $SR$  and

$Z$ . The full weight can be decomposed into a component that changes the distribution of  $Z$  and a component that changes the conditional distribution of  $SR$  given  $Z$ . Define

$$\omega_Z(z) = \frac{dF_1(z)}{dF_0(z)}.$$

The sex-ratio-specific reweighting factor is

$$\omega_{SR}(x) = \frac{\omega(x)}{\omega_Z(z)} = \frac{f_1(SR | Z)}{f_0(SR | Z)}.$$

This weight assigns the 1975 cohort the 1990 distribution of sex-ratio exposure conditional on education, while holding the marginal education distribution at its 1975 level.

The three means used in the decomposition are

$$\begin{aligned}\mu_0 &= \int y f_0(y) dy, \\ \mu_1 &= \int y f_0(y | x) \omega(x) dF_0(x) dy, \\ \mu_2 &= \int y f_0(y | x) \omega_{SR}(x) dF_0(x) dy.\end{aligned}$$

Here,  $\mu_0$  is the observed 1975 mean,  $\mu_1$  gives the 1975 cohort the 1990 distributions of both education and sex-ratio exposure, and  $\mu_2$  changes only the sex-ratio distribution. The total composition effect is

$$\mu_1 - \mu_0 = \underbrace{\mu_2 - \mu_0}_{\text{sex-ratio contribution}} + \underbrace{\mu_1 - \mu_2}_{\text{education contribution}}.$$

I report each component as a share of the total change in predicted age at first marriage. The decomposition is estimated separately by gender and by urban/rural status.

## References

- Abramitzky, Ran, Adeline Delavande, and Luis Vasconcelos. 2011. “Marrying Up: The Role of Sex Ratio in Assortative Matching.” *American Economic Journal: Applied Economics* 3 (3): 124–57.
- Angrist, Josh. 2002. “How Do Sex Ratios Affect Marriage and Labor Markets? Evidence from America’s Second Generation.” *Quarterly Journal of Economics* 117 (3): 997–1038.
- Becker, Gary S. 1973. “A Theory of Marriage: Part I.” *Journal of Political Economy* 81 (4): 813–46.
- Chen, Ruoyu, and Lingxiang Zhang. 2019. “Imbalance in China’s Sex Ratio at Birth: A Review.” *Journal of Economic Surveys* 33 (3): 1050–69.
- Chen, Yuyu, Hongbin Li, and Lingsheng Meng. 2013. “Prenatal Sex Selection and Missing Girls in China: Evidence from the Diffusion of Diagnostic Ultrasound.” *Journal of Human Resources* 48 (1): 36–70. doi: 10.3368/jhr.48.1.36.
- CNDeepData. 2023. “CNDD Population Census Microdata Research Data Collection.” Data set. <https://cndeepdata.pro/2023/08/14/cn-dd-%E4%BA%BA%E5%8F%A3%E6%99%AE%E6%9F%A5%E5%BE%AE%E8%A7%82%E7%A0%94%E7%A9%B6%E6%95%B0%E6%8D%AE%E5%90%88%E9%9B%86/>.
- Chiappori, Pierre-André, Murat Iyigun, and Yoram Weiss. 2009. “Investment in Schooling and the Marriage Market.” *American Economic Review* 99 (5): 1689–1713.
- Dong, Hao, and Yu Xie. 2023. “Trends in Educational Assortative Marriage in China over the Past Century.” *Demography* 60 (1): 123–45.
- Ebenstein, Avraham. 2010. “The ‘Missing Girls’ of China and the Unintended Consequences of the One Child Policy.” *Journal of Human Resources* 45 (1): 87–115. doi: 10.3368/jhr.45.1.87.
- Golley, Jane, and Rodney Tyers. 2012. “China’s Gender Imbalance and Its Economic Performance.” Working paper.
- Goodkind, Daniel. 2011. “Child Underreporting, Fertility, and Sex Ratio Imbalance in China.” *Demography* 48: 291–316.



- Greenwood, Jeremy, Nezih Guner, Georgi Kocharkov, and Cezar Santos. 2014. “Marry Your Like: Assortative Mating and Income Inequality.” *American Economic Review* 104 (5): 348–53.
- Guilmoto, Christophe Z. 2012. “Skewed Sex Ratios at Birth and Future Marriage Squeeze in China and India, 2005–2100.” *Demography* 49 (1): 77–100.
- Guo, Lijia, Jiashun Huang, and You Zhang. 2019. “Education Development in China: Education Return, Quality, and Equity.” *Sustainability* 11 (13): 3750.
- Iyigun, Murat, and P. Randall Walsh. 2007. “Building the Family Nest: Premarital Investments, Marriage Markets, and Spousal Allocations.” *Review of Economic Studies* 74 (2): 507–35.
- Jingguan Zhijia. 2021. “1990 Population Census 1 Percent Sample Microdata, SPSS and Stata Version.” Online forum post. <https://bbs.pinggu.org/thread-10611646-1-1.html>.
- Lafortune, Jeanne. 2013. “Making Yourself Attractive: Pre-Marital Investments and the Returns to Education in the Marriage Market.” *American Economic Journal: Applied Economics* 5 (2): 151–78.
- Li, Bingjing, and Hongliang Zhang. 2017. “Does Population Control Lead to Better Child Quality? Evidence from China’s One-Child Policy Enforcement.” *Journal of Comparative Economics* 45 (2): 246–60.
- National Bureau of Statistics of China. 1999. “Provisions on Statistical Division of Urban and Rural Areas (Trial).” National Bureau of Statistics of China. <https://www.stats.gov.cn/sj/pcsj/rkpc/5rp/html/append7.htm>.
- National Bureau of Statistics of China. 2008. “Provisions on Statistical Division of Urban and Rural Areas.” National Bureau of Statistics of China. [https://www.stats.gov.cn/sj/tjbz/gjtjbz/202302/t20230213\\_1902742.html](https://www.stats.gov.cn/sj/tjbz/gjtjbz/202302/t20230213_1902742.html).
- National Bureau of Statistics of China. 2009. “Rules for Compiling Statistical Zoning Codes and Urban-Rural Classification Codes.” National Bureau of Statistics of China. [https://www.stats.gov.cn/sj/tjbz/gjtjbz/202302/t20230213\\_1902741.html](https://www.stats.gov.cn/sj/tjbz/gjtjbz/202302/t20230213_1902741.html).
- National Bureau of Statistics of China. 2025. “How Are Urban and Rural Areas Statistically Classified?” National Bureau of Statistics of China. [https://www.stats.gov.cn/zt/tjws/tjbz/202301/t20230101\\_1903381.html](https://www.stats.gov.cn/zt/tjws/tjbz/202301/t20230101_1903381.html).

- National Bureau of Statistics of China and IPUMS International. 1990. “China Fourth National Population Census 1990: IPUMS Subset.” Data set. <https://microdata.worldbank.org/index.php/catalog/462>.
- Rosenzweig, Mark R., and Junsen Zhang. 2009. “Do Population Control Policies Induce More Human Capital Investment? Twins, Birth Weight, and China’s One-Child Policy.” *Review of Economic Studies* 76 (3): 1149–74.
- Wei, Shang-Jin, and Xiaobo Zhang. 2011. “The Competitive Saving Motive: Evidence from Rising Sex Ratios and Savings Rates in China.” *Journal of Political Economy* 119 (3): 511–64.
- Wu, Xiaogang. 2010. “Economic Transition, School Expansion, and Educational Inequality in China, 1990–2000.” *Research in Social Stratification and Mobility* 28 (1): 91–108.
- Wu, Ruijun, Qixin Xue, and Zhihua Luo. 2022. “Turning Changes in China’s Population Migration and Urbanization Pattern: 2000–2020.” *The Journal of Shanghai Administration Institute* 23 (1): 74–86. <http://hprc.cssn.cn/gsyj/jjs/jgzhn/202207/P020220725541375410796.pdf>.
- Zhang, Junsen. 2017. “The Evolution of China’s One-Child Policy and Its Effects on Family Outcomes.” *Journal of Economic Perspectives* 31 (1): 141–60.
- Zheng, Zhenzhen, and Ge Yang. 2013. “Current Status and Future Trends of Population Mobility in China.” People’s Daily Online–People’s Tribune. Original article in Chinese. <http://theory.people.com.cn/n/2013/0514/c112851-21473033.html>.
- Zhou, Shikai, and Sangui Wang. 2024. “Gender Imbalance in the Marriage Market and Housing Demand: Evidence from China.” *Sustainability* 16 (14): 5861.